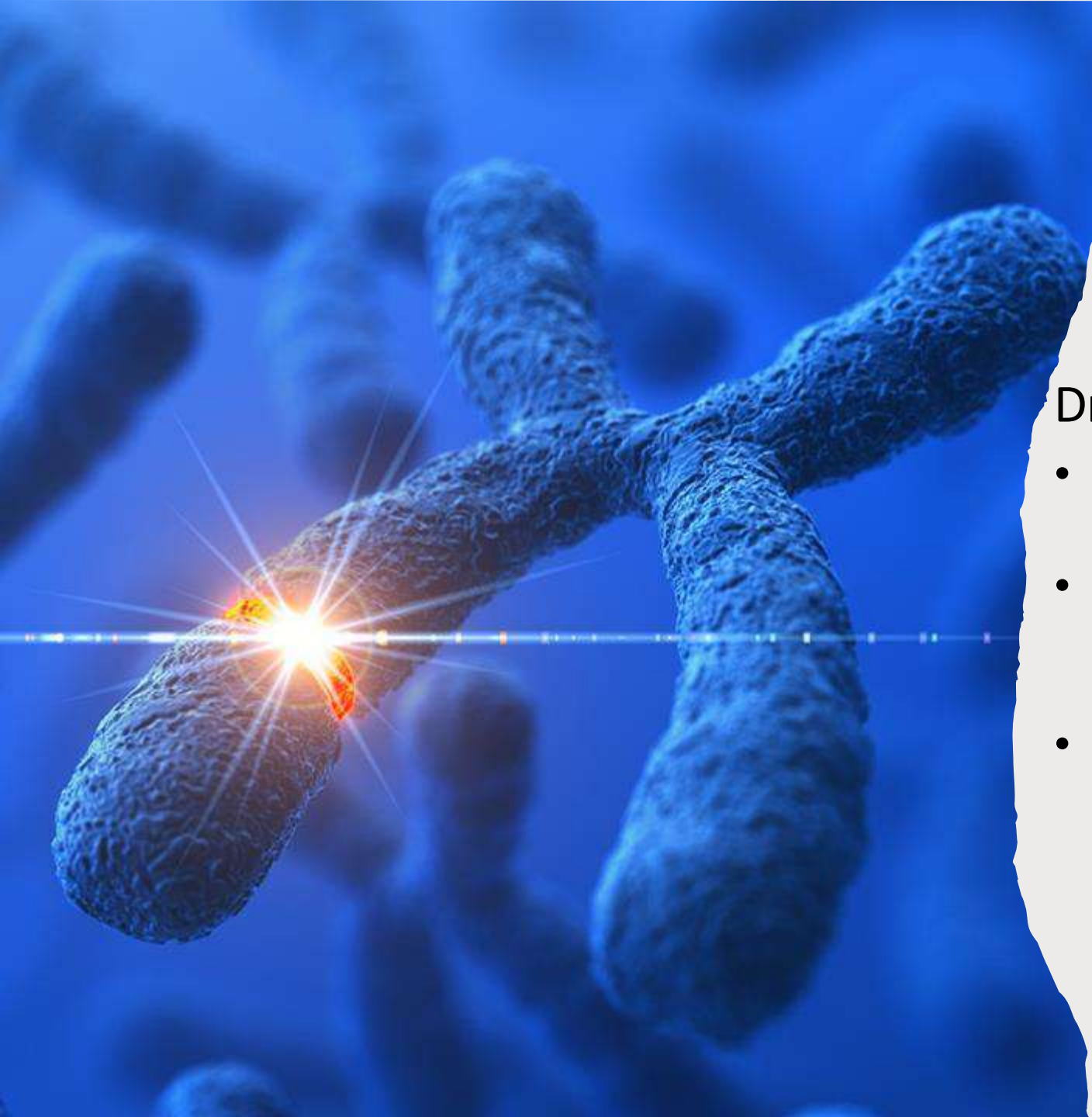




Genetic Variants

(Chapter 2, page 19-24)

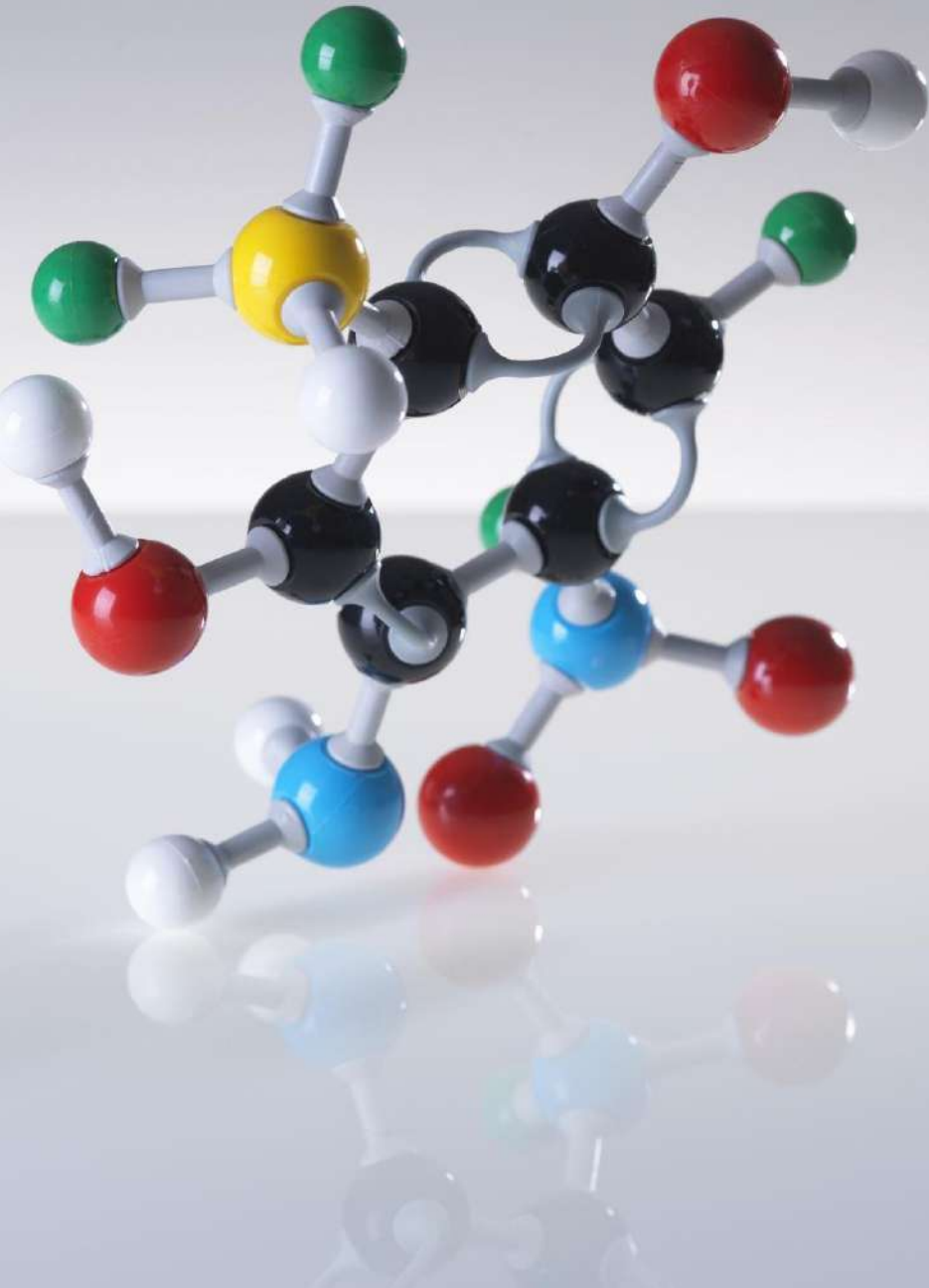
Dr. Deema Aljeaid

- MBBS, FAAP, FACMG
- Consultant Clinical Geneticist/Assistant Professor
- Department of Genetic Medicine, Faculty of Medicine, KAU

Dr. med. Wisam Habhab

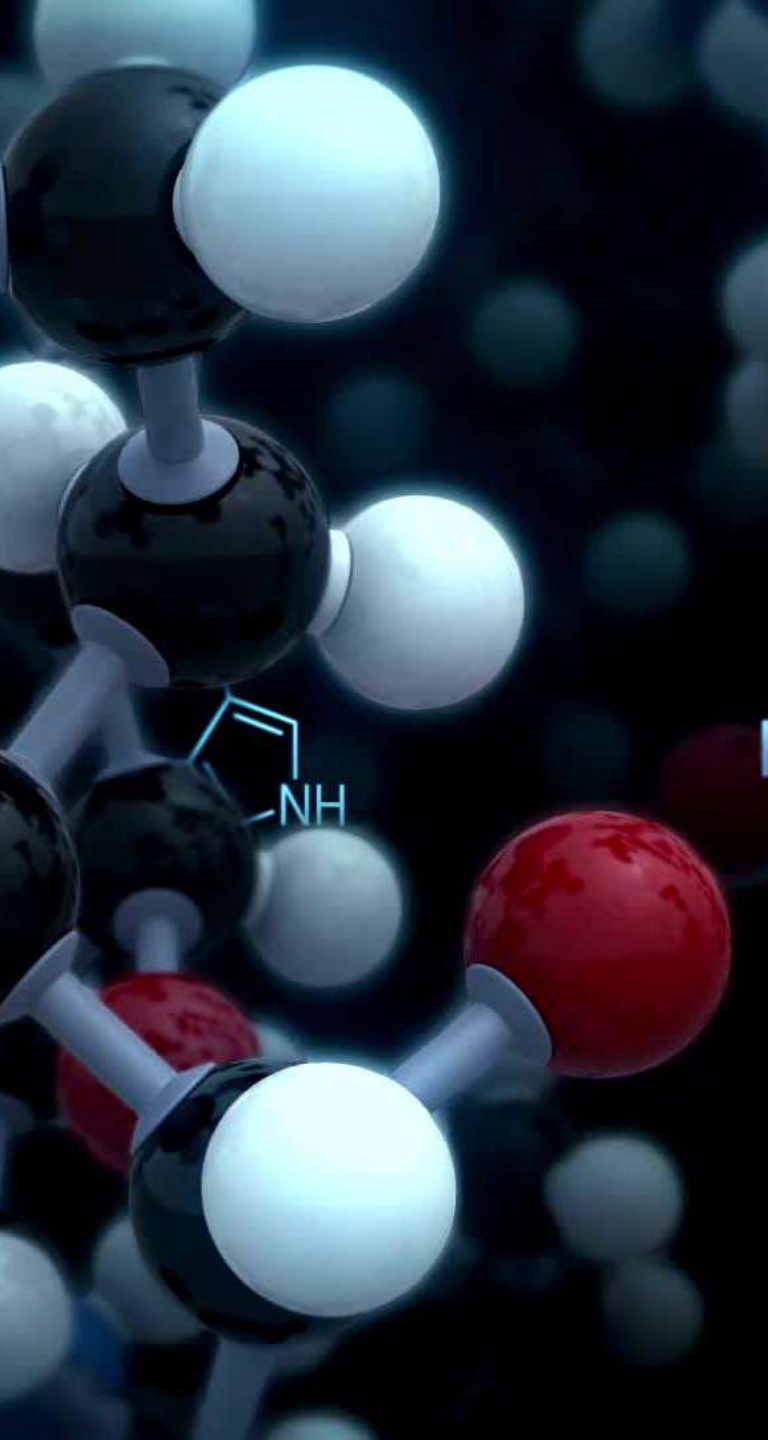
- Consultant Clinical Geneticist/Assistant Professor
- Department of Genetic Medicine, Faculty of Medicine, KAU

2nd year medicine, 2026



Learning Outcomes

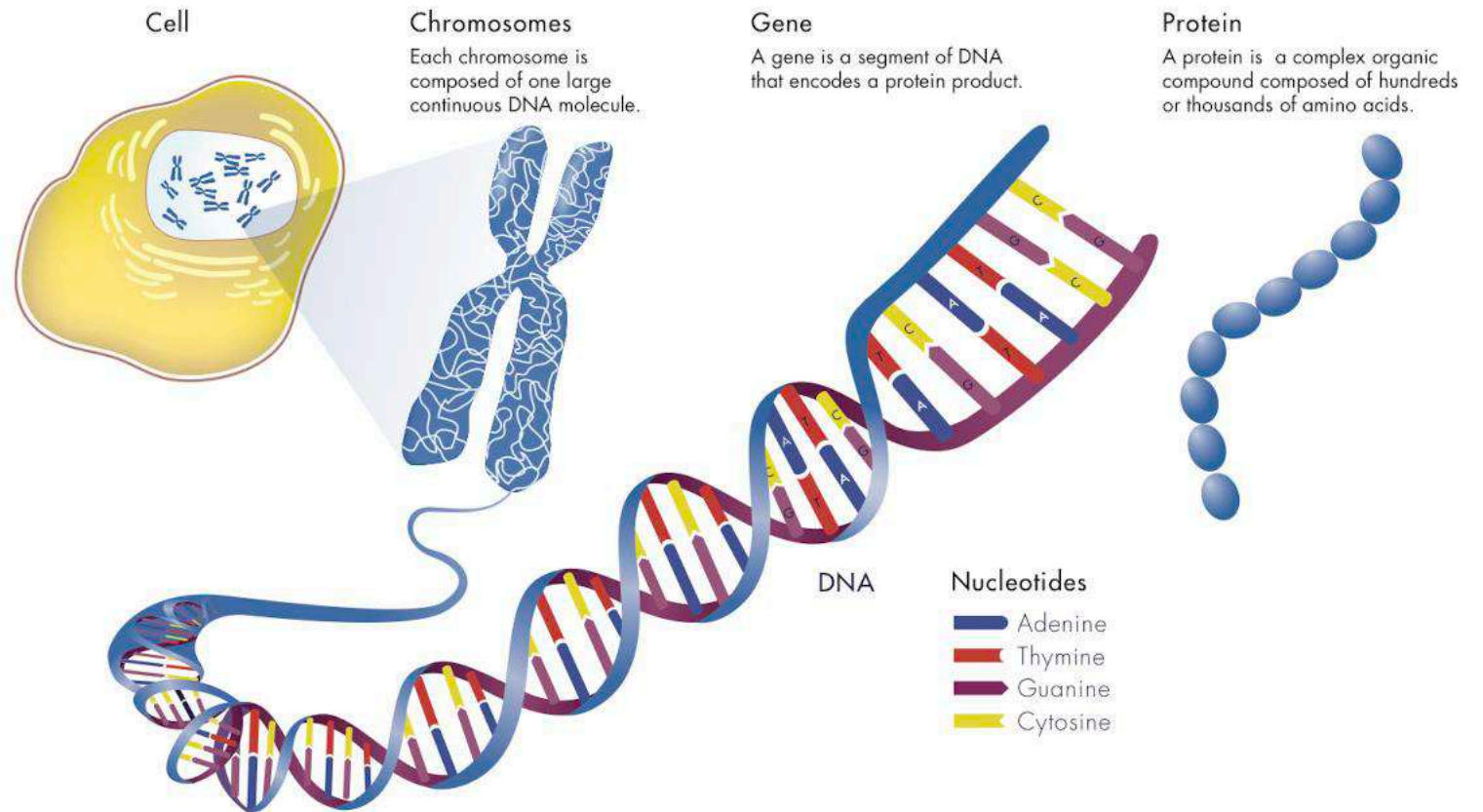
- Define genetic variants/mutations
- Classify types of variants
- List the causes of variants
- Understand the nomenclature of genetic variation
- Give examples of clinical consequences of variants
- Recognize variants by inheritance:
 - Homozygous and compound heterozygous variants in autosomal recessive disorders
 - De novo and inherited variants



OUTLINE

- Definition
- Causes of mutations/variants
 - Spontaneous and induced
- Germline vs somatic variants
- Type of variants:
 - SNV/substitutions, Indels, and SVs
- Structural effect of variants on the protein:
 - Synonymous (silent)
 - Non-synonymous (missense, nonsense and frameshift)
- Variants in non-coding DNA:
 - splicing variants
- Functional effects of variants on the protein:
 - Loss-of-function, gain-of-function, and dominant-negative variants
- Variant by inheritance
 - Homozygous, compound heterozygous, & *de novo*
- Genotype-phenotype correlation
- Nomenclature of genetic variation
- Example of clinical consequences of genetic mutation

Chromosome to Gene to Protein

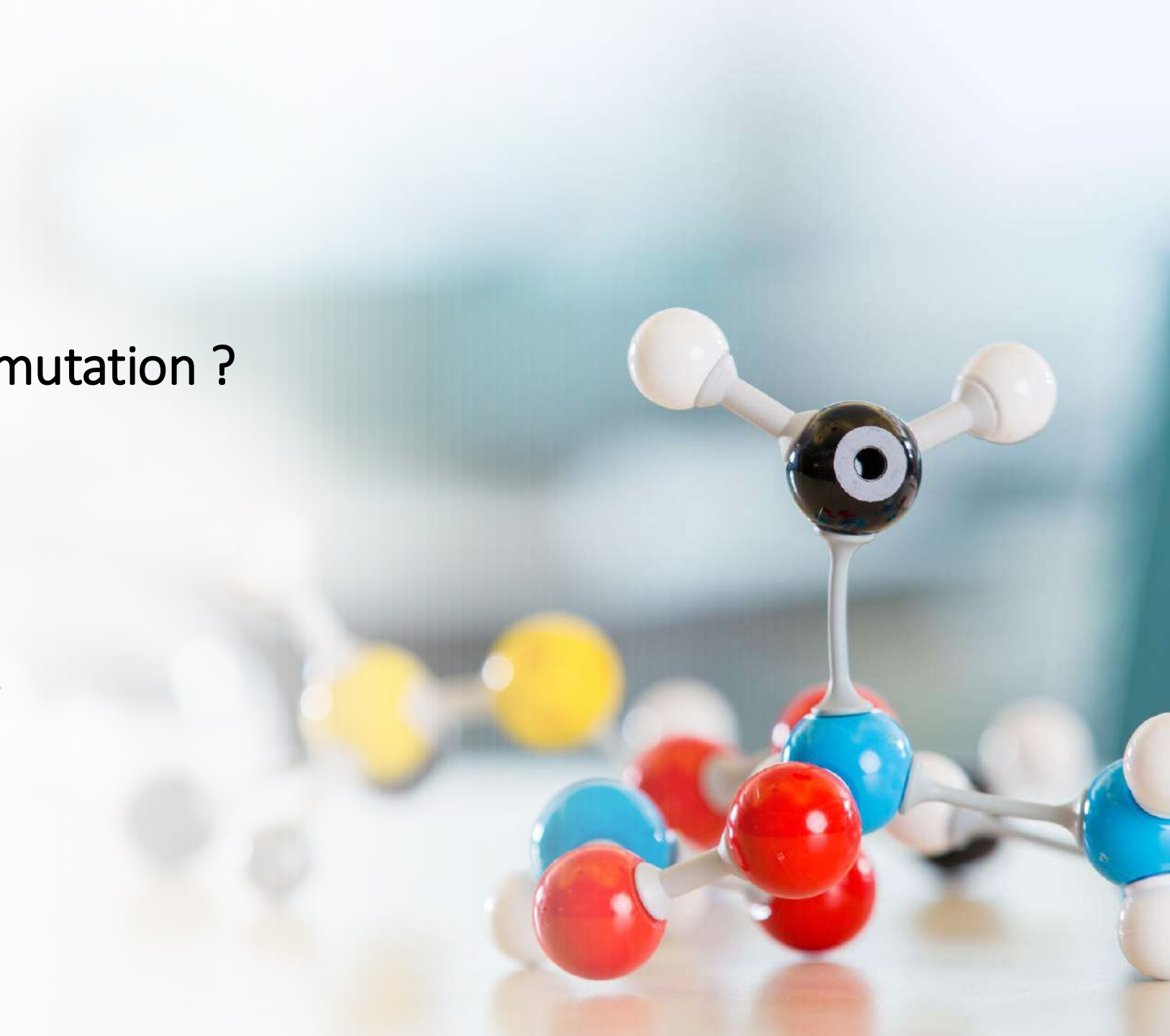




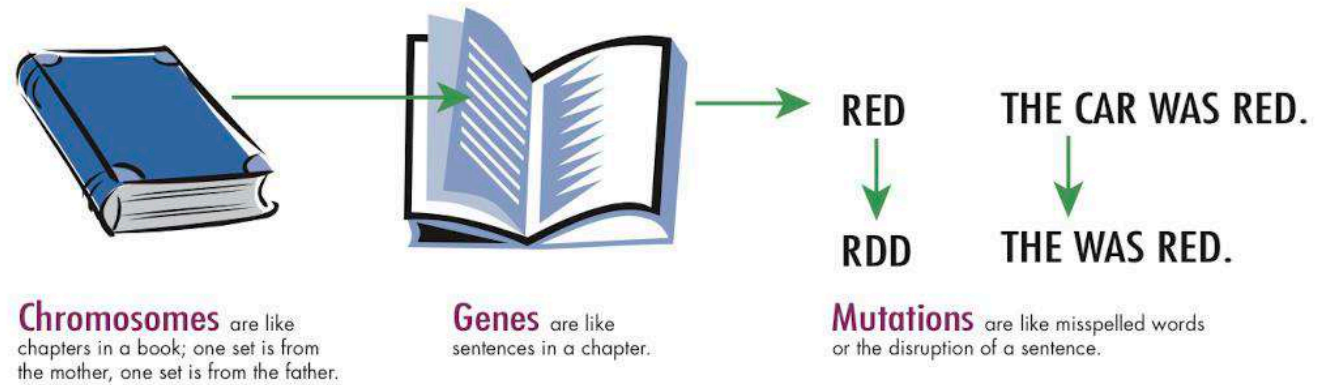
What is a genetic variant/mutation ?

And

Why are they important ?



Definition



- **Gene variants**: an alteration or change in the genetic material (the genome), (i.e., change in the DNA sequence of a gene)
- Previously known as 'mutations,' the term '**variant**' is now considered more accurate, **as changes in DNA do not always cause disease.**
- The human genome contains approximately 4 million variants

Causes of Variants

Spontaneous:

- Naturally occurring variants are referred to as spontaneous variants (the majority)
- Thought to arise through chance errors in chromosomal division, DNA replication or repair

Induced:

- By exposure to mutagens (which are environmental agents that cause variants)
- These include natural or artificial ionizing radiation and chemical or physical mutagen
- Example of chemical mutagens: formaldehyde, benzene and food additives

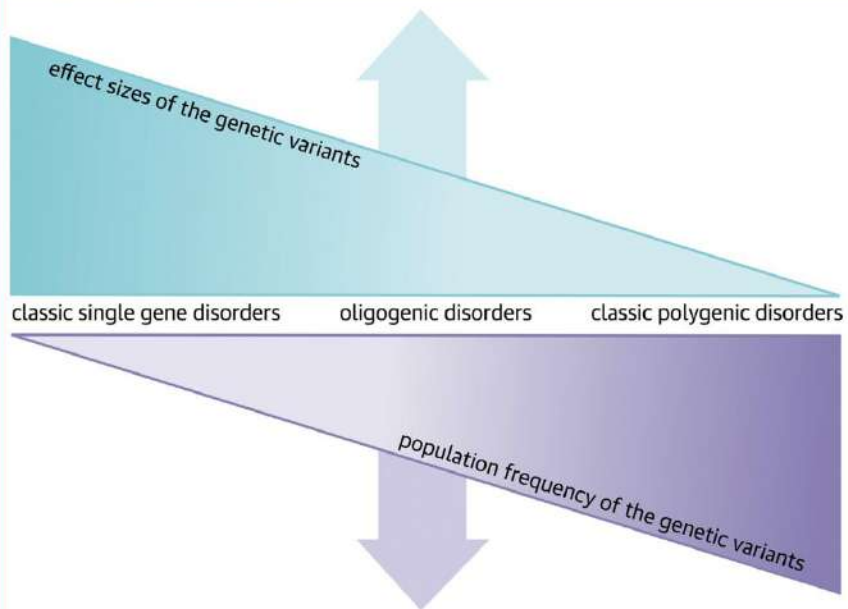
Impact of Genetic Variations

Variant Classification



- A variant of uncertain significance (VUS) is a change in the DNA/genetic material, that may or may not be related to the patient's concern and reason for testing.

CENTRAL ILLUSTRATION Genetic Disorders as a Continuum



Marian, A.J. J Am Coll Cardiol Basic Trans Science. 2020;5(10):1029-42.

Genetic variants impart a gradient of functional and clinical effect sizes that vary from negligible to very large. Population frequency of the genetic variants inversely correlates with their effect sizes. Those with large effect sizes are rare and those with small effect sizes are common. The classic single-gene disorders result from the very large effects of rare variants, whereas the polygenic disorders result from the cumulative effects of multiple variants with small effect sizes. Most genetic disorders are in between the 2 ends of the spectrum.

- Genetic variation contributes to the evolution of species.
- Natural selection favors beneficial traits, allowing them to survive and be inherited.
- Variants may be benign, pathogenic "disease-causing", or of unknown significance.
- Most variants have no health consequences, some contribute to individual differences (eye color, hair color, etc.), and other may cause or increase the risk of certain disorders.

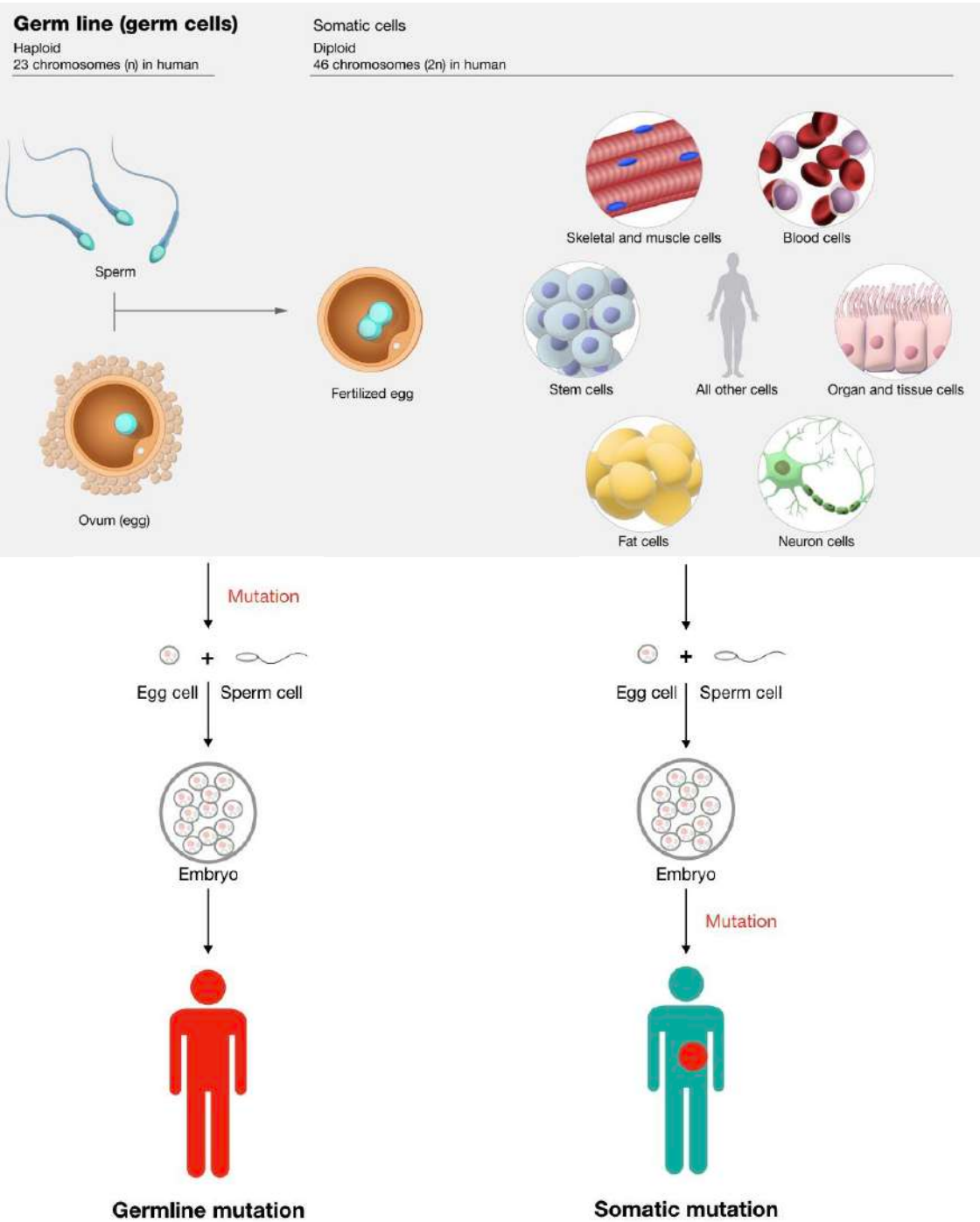
Classification based on
cell type

```
graph TD; A[Classification based on cell type] --> B[Germline variants]; A --> C[Somatic variants];
```

Germline variants

Somatic
variants

Germ-line variant (in gonadal tissue or a gamete)	Somatic variant
-Occurs in germline cell (gonadal tissue or a gamete “egg” or “sperm”)	-Occurs in somatic cells
- Can be transmitted to future generations (inherited)	- Cannot be transmitted to offspring (not inherited)
- Inherited from parents	- Acquired
- Will be found in all body cells of the new organism	- The effect is local affecting certain part of the body
- Can cause early-onset disease	- Can cause adult-onset disease
e.g. Sickle cell disease	e.g. Cancer



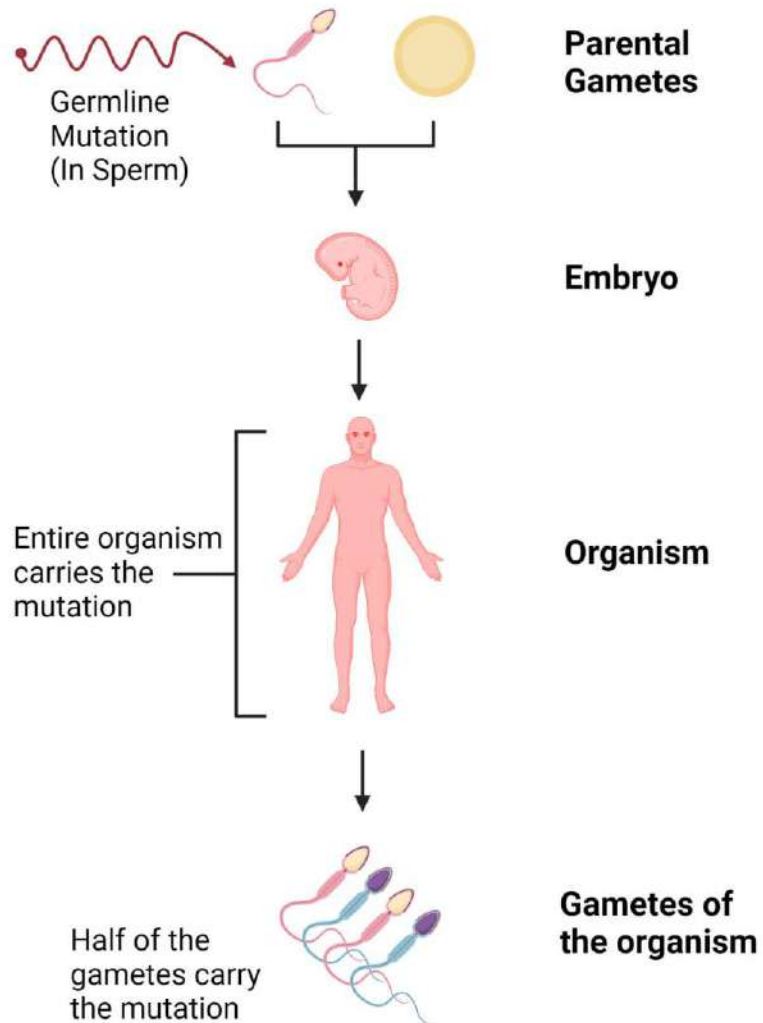


Figure: Germline Mutation

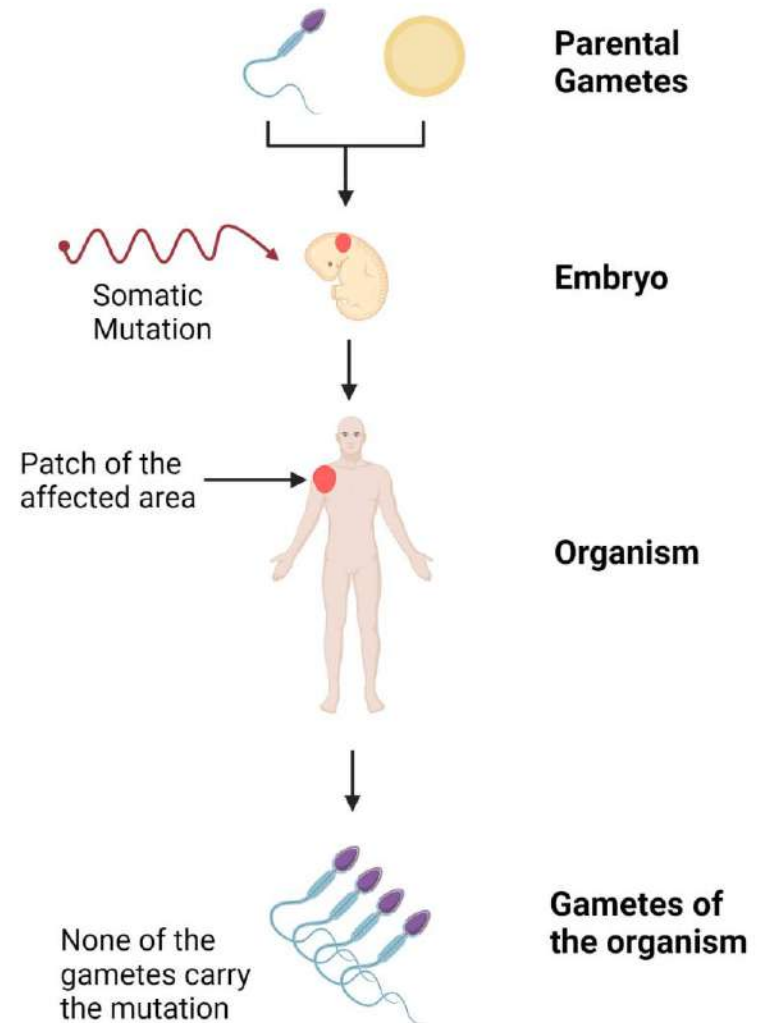
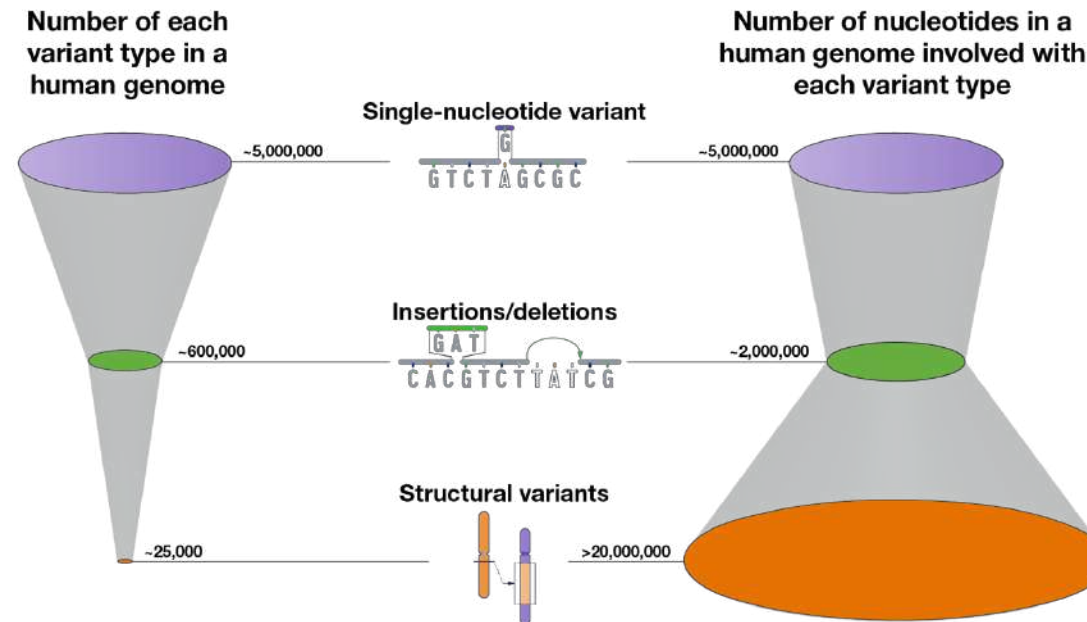
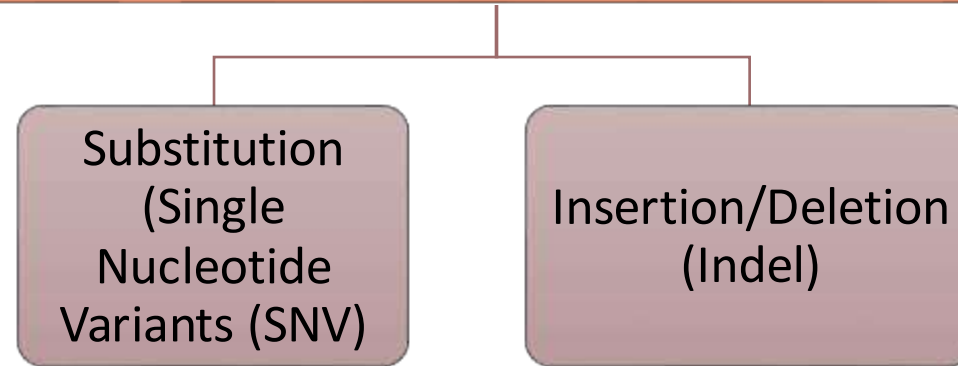


Figure: Somatic Mutation

Classification based on type of alteration

(Types/Classes of Variants)



Single Nucleotide Variants (SNV) or Substitutions

- Known as point mutations/variants
- The replacement of a single nucleotide by another
- Most common type

- **Transition:**

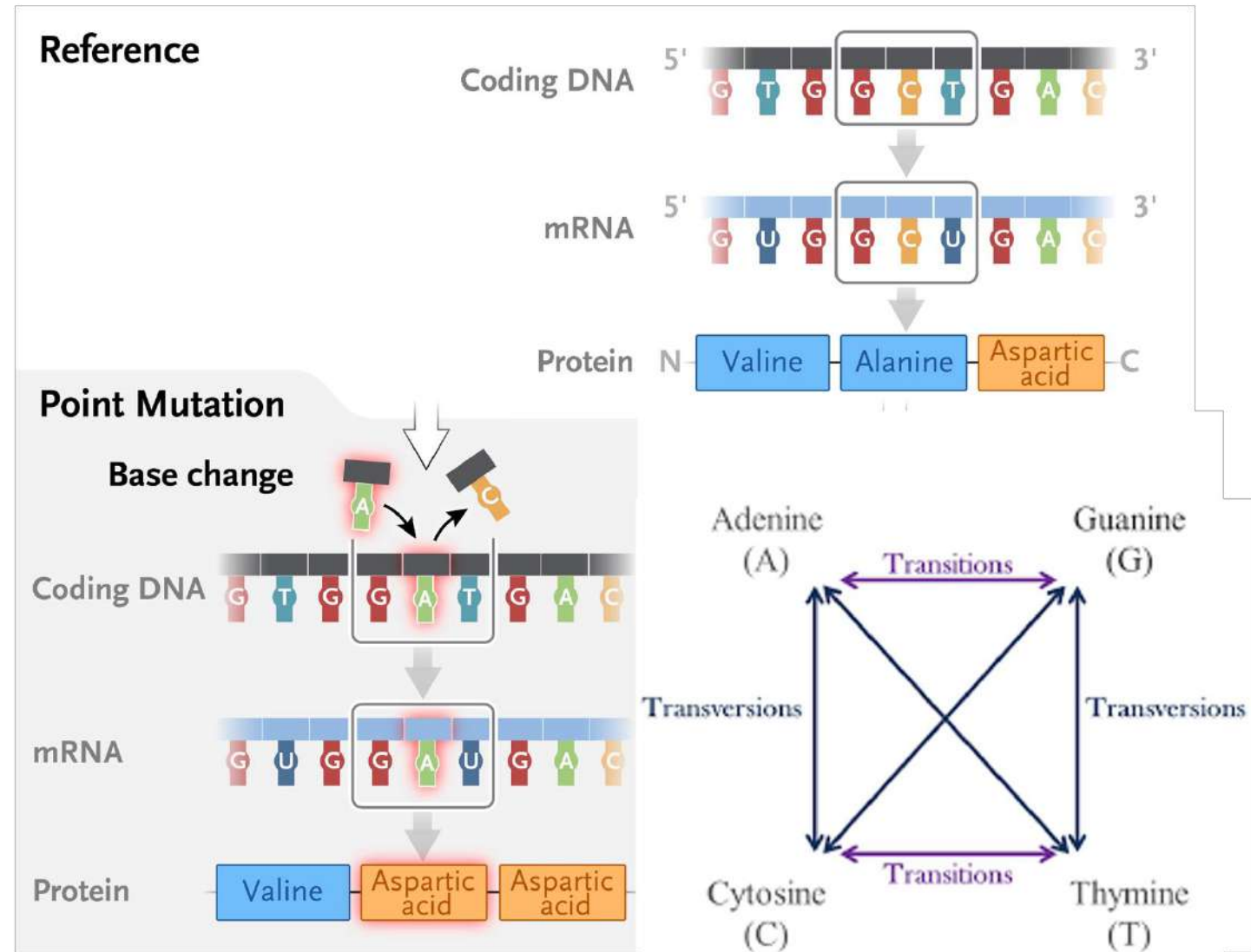
Purine is substituted with another **purine (A,G)** or **pyrimidine** is substituted with another **pyrimidine (C,T)**

- **Transversion:**

Purine is substituted for a **pyrimidine**, or a **pyrimidine**

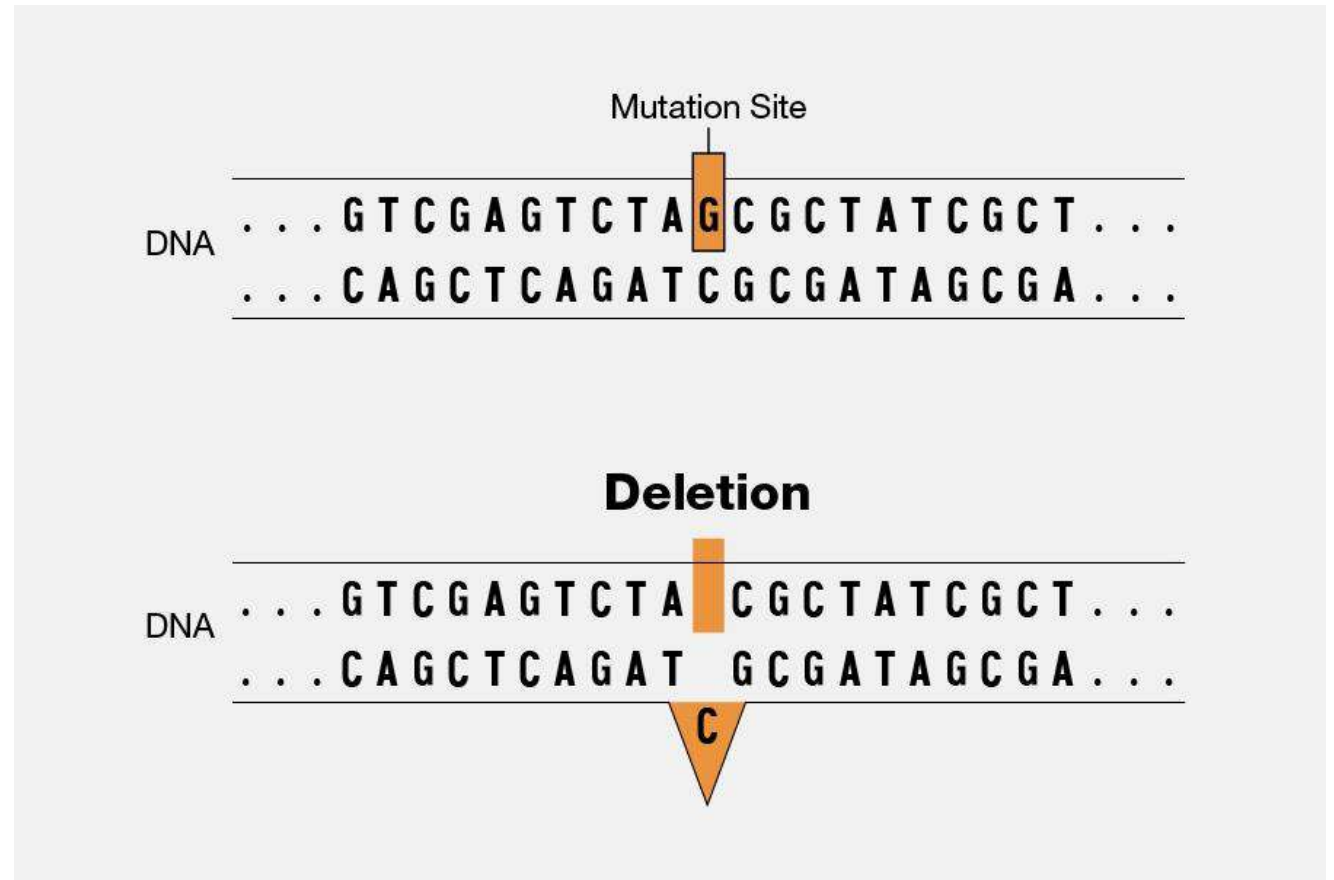
replaces a **purine**

- Transitions occur more frequently than transversions.



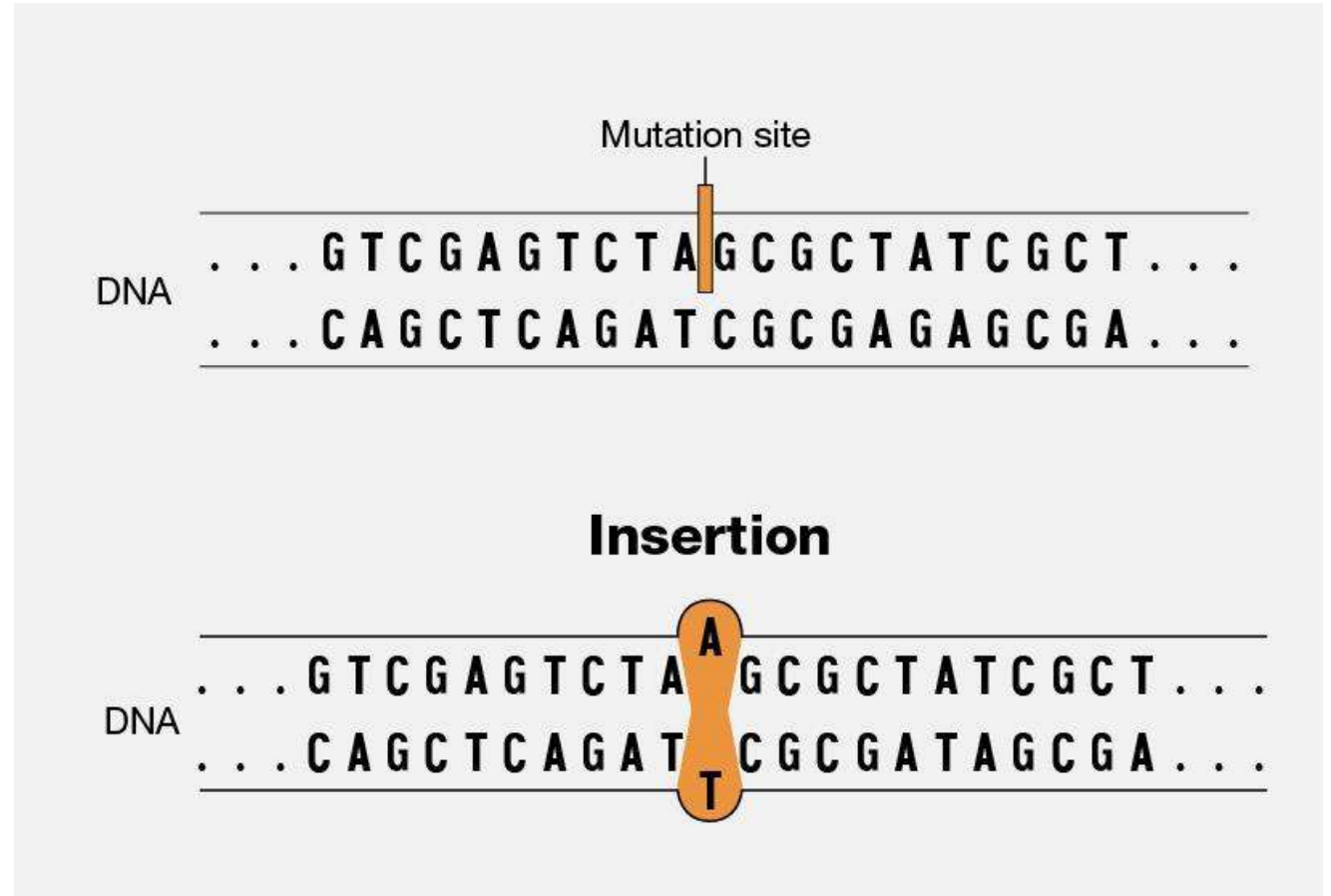
Base Deletions

- The **loss** of one or more nucleotide
- If occurs in coding sequences and involves one, two or more nucleotides that are **not a multiple of three, the reading frame will be disrupted**
- Larger deletions may result in partial or whole gene deletions → loss of gene function or expression
- May arise through unequal cross over between repeat sequences



Base Insertions

- The **addition** of one or more nucleotides
- If occurs in coding sequences and involves one, two or more nucleotides that are **not a multiple of three, the reading frame will be disrupted**
- May arise through unequal cross over between repeat sequences



**Structural Effects of Variants on the Protein
(variants/mutations in the coding DNA)**

```
graph TD; A["Structural Effects of Variants on the Protein  
(variants/mutations in the coding DNA)"] --> B["Synonymous"]; A --> C["Non-synonymous"]; B --> D["Silent variants"]; C --> E["Missense variants"]; C --> F["Nonsense variants"]; C --> G["Frameshift variants"];
```

Synonymous

Non-synonymous

Silent variants

Missense variants

Nonsense variants

Frameshift
variants

THE CAT SAW THE DOG

Change of
one letter

Substitution

THE **B**AT SAW THE DOG
THE CAT SAW THE **H**OG
THE CAT SA**T** THE DOG

Point mutation

Missense

Loss of
one letter

Deletion

THE ATS AWT HED OG

Loss of C

Frameshift mutation

Gain of
one letter

Insertion

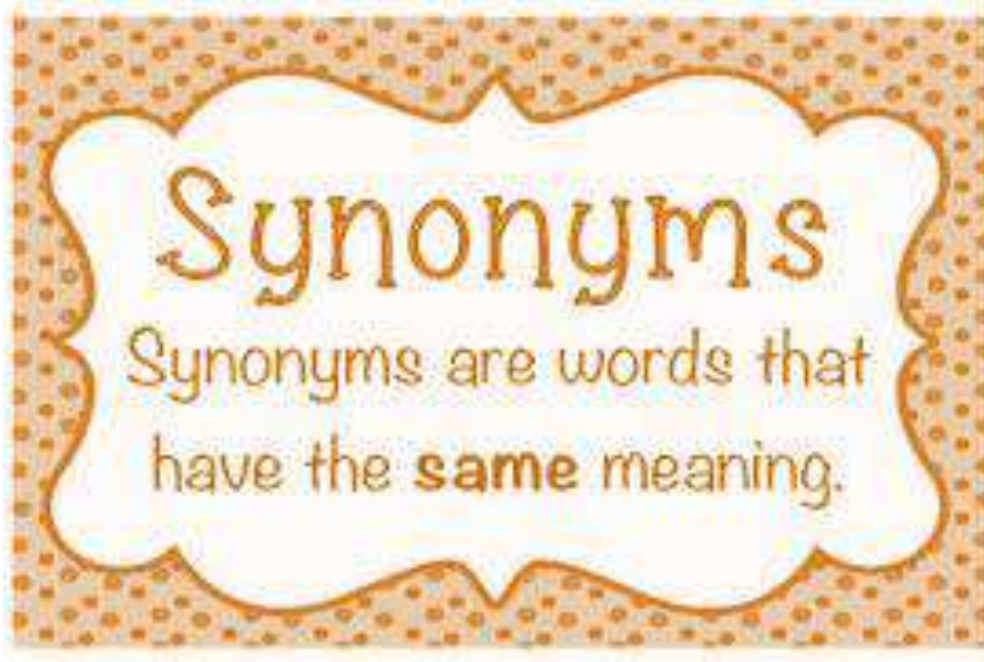
THE **C****M**A TSA WTH EDO G

Insertion of M

Frameshift mutation

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Synonymous or Silent Variant



- A variant does **NOT** alter the polypeptide product of the gene (i.e., **NO** change to the amino acid sequence)
- A single base-pair substitution, particularly if it occurs in the third position of a codon because of the degeneracy of the genetic code
→ will often result in another triplet that codes for the **same amino acid** with **NO** alteration in the properties of the resulting protein

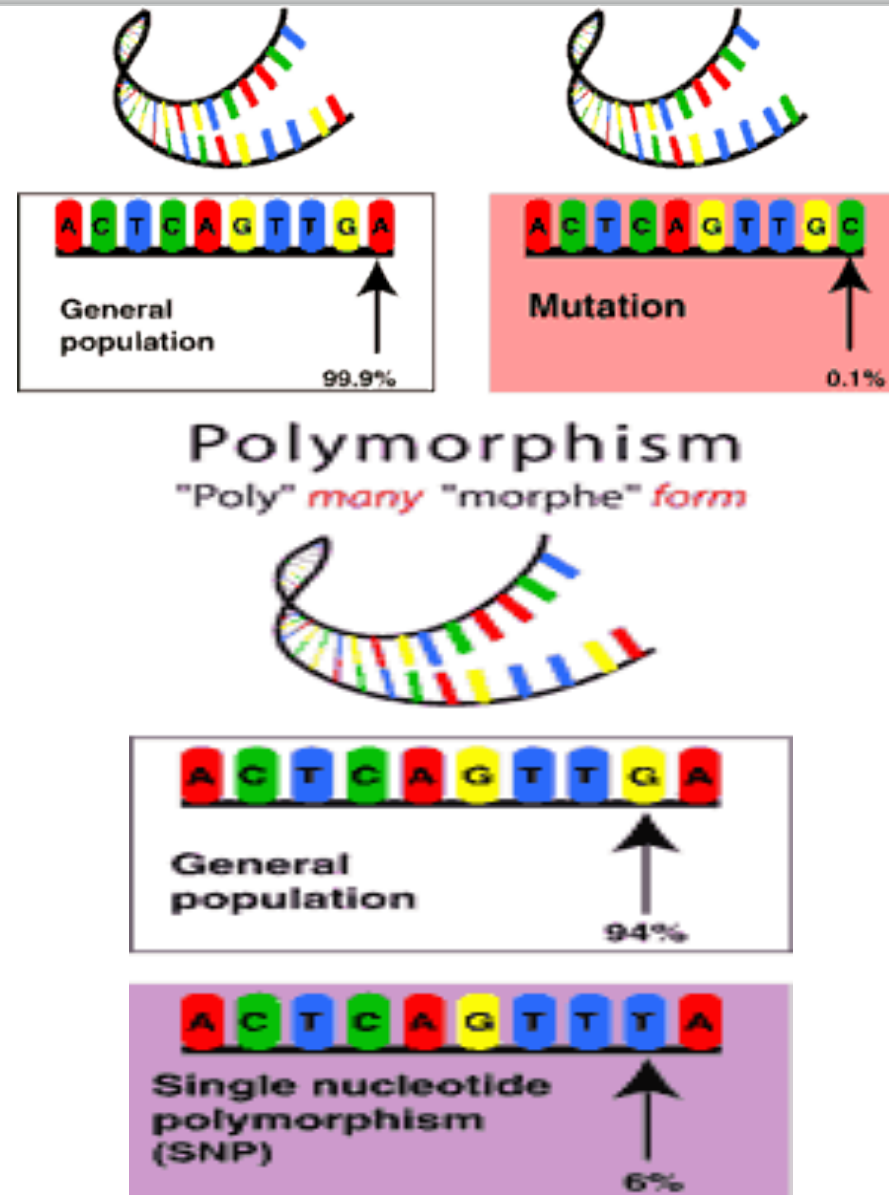


UUU } UUC } Phe UUA } UUG } Leu	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA } Stop UAG }	UGU } Cys UGC } UGA } Stop UGG } Trp
CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } Arg CGC } CGA } CGG }
AUU } AUC } Ile AUA } AUG } Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }
GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } Gly GGC } GGA } GGG }

Polymorphism

Polymorphisms:

- Sequence variants with no obvious effect on phenotype
- SNP is a single base substitution, occurring in more >1% of the population
- SNPs are the most common type of variation in the human genome



Single nucleotide polymorphism (SNP)

Individual 1

Maternal ...CGATATTCC**T**ATCGAATGTC...

Paternal ...CGATATTCC**C**ATCGAATGTC...

Individual 2

Maternal ...CGATATTCC**C**ATCGAATGTC...

Paternal ...CGATATTCC**C**ATCGAATGTC...

Short tandem repeat polymorphism (STRP)

Individual 3

Maternal ...CGATATTCC**CAGCAGCAG**ATCGAATGTC...

Paternal ...CGATATTCC**CAGCAGCAGCAGCAG**ATCGAATGTC...

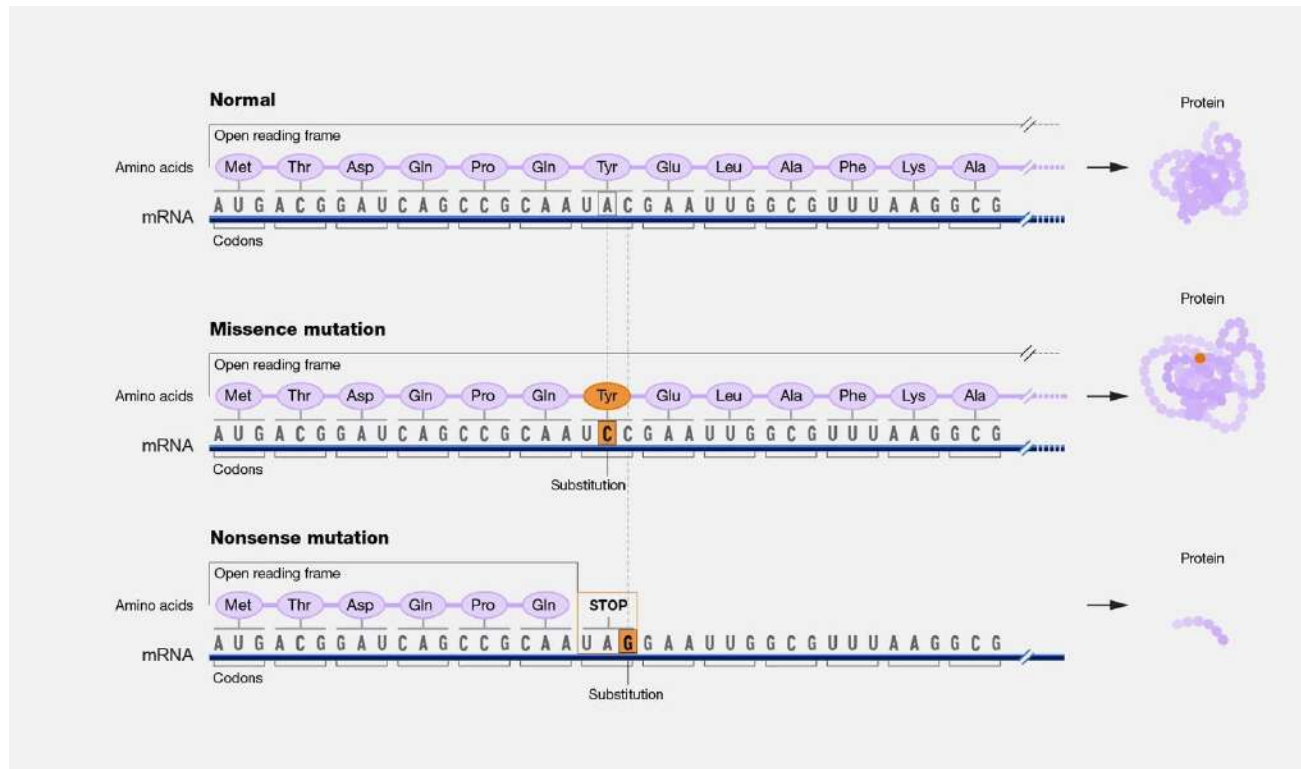
Individual 4

Maternal ...CGATATTCC**CAGCAGCAGCAGCAG**ATCGAATGTC...

Paternal ...CGATATTCC**CAGCAGCAGCAGCAGCAGCAGCAG**ATCGAATGTC...

Polymorphism

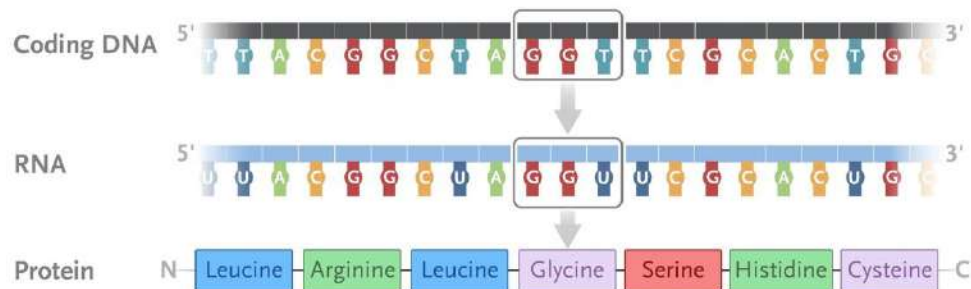
Non-Synonymous



- If a variant leads to alteration in the encoded polypeptide product of the gene
- **Less frequent** than synonymous mutations
- Result in abnormal function of the gene product
- Usually associated with disease, or lethality
- Can occur in one of three main ways: missense, nonsense, frameshift

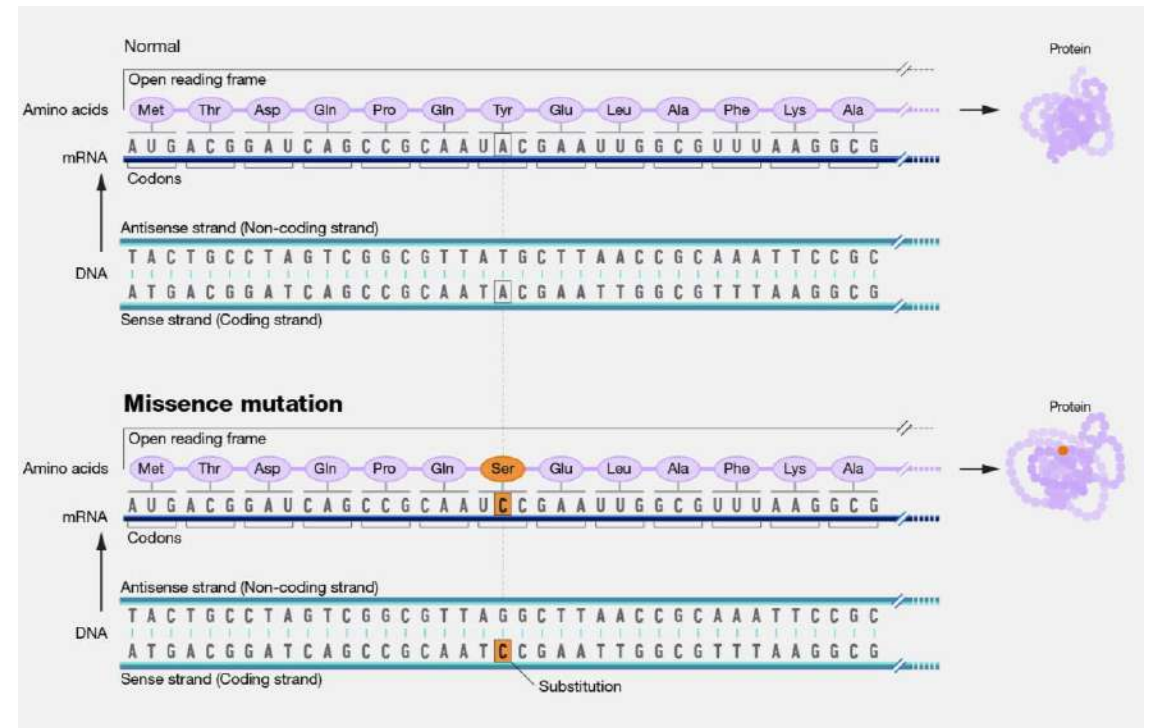
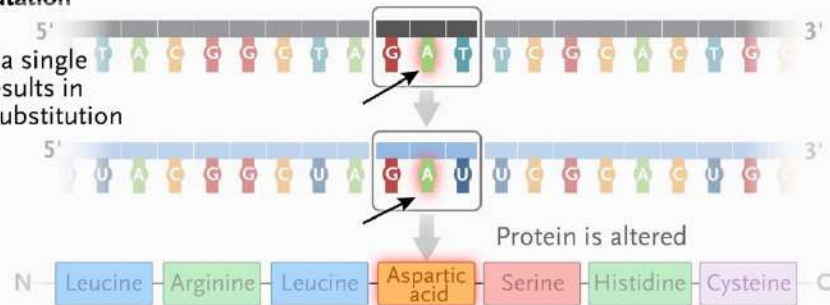
Missense Variant/Mutation

- Result in coding for a different amino acid and the synthesis of an altered protein
- Conservative vs non-conservative



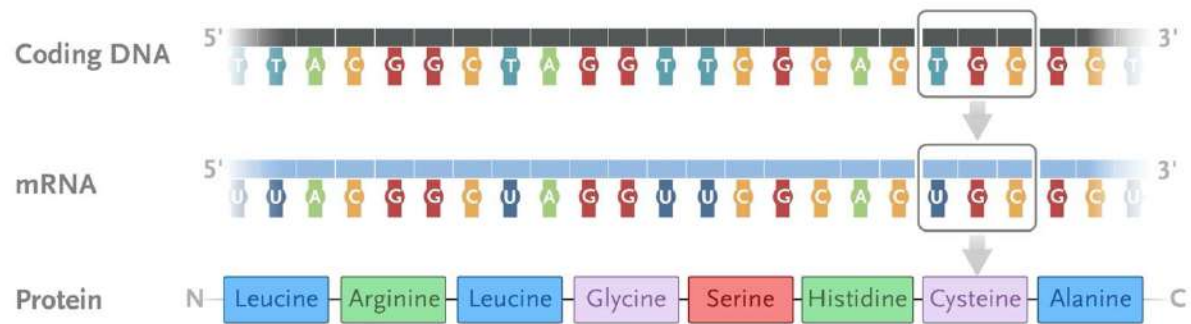
Missense mutation

Alteration of a single nucleotide results in amino acid substitution



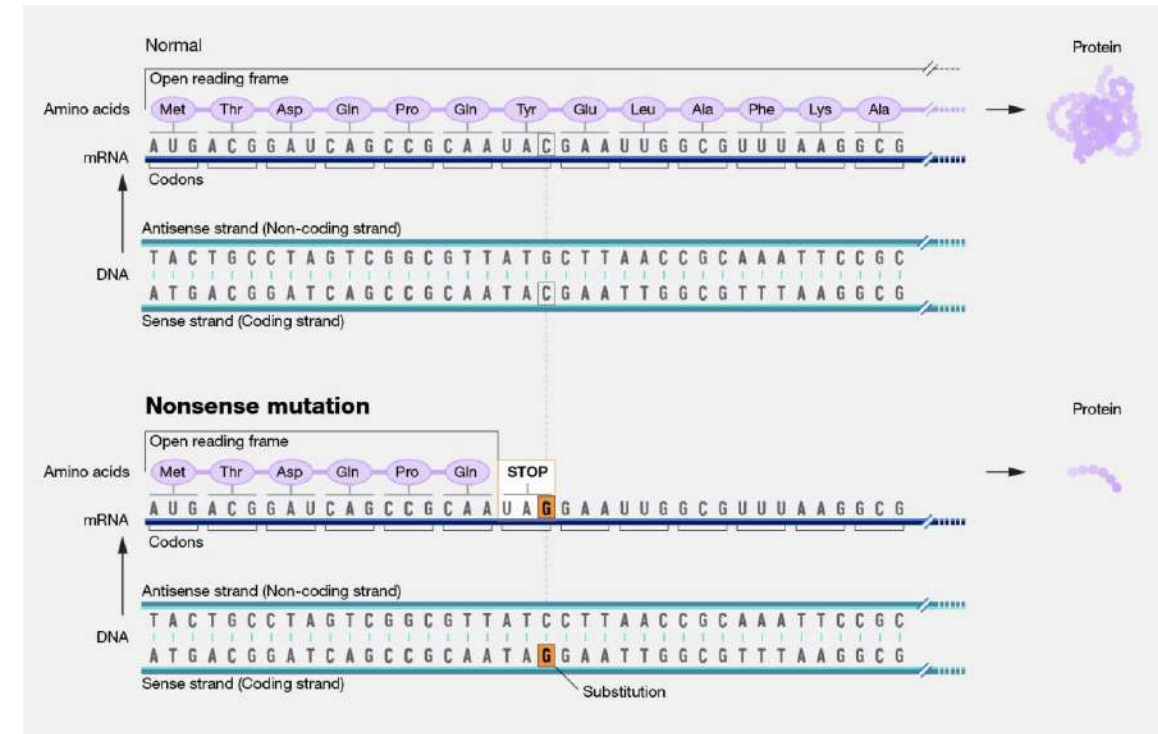
Nonsense(Stopgain) Variant/Mutation

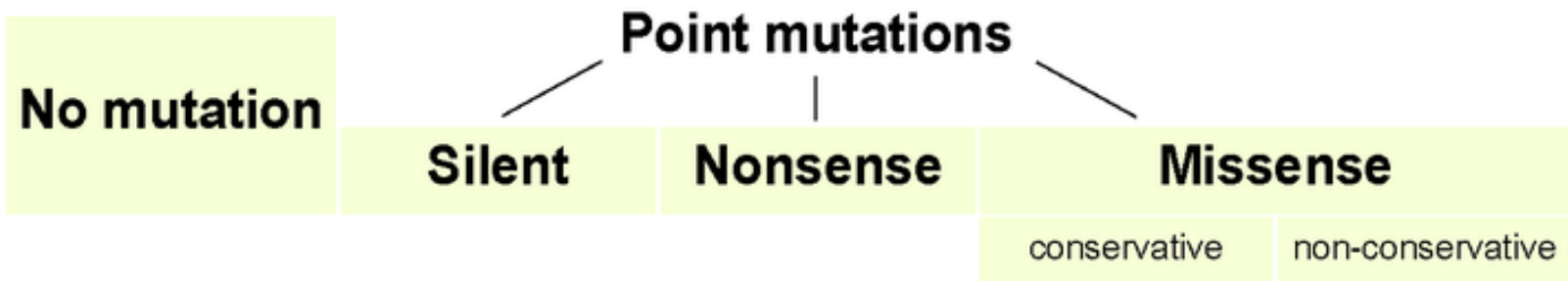
- A substitution that leads to the generation of one of the stop codons (**UAA, UAG, or UGA**)
- Will result in a premature termination of translation of a peptide chain (truncated protein)
- Nonsense-mediated decay eliminates mRNA with premature stop signals, preventing harmful truncated proteins

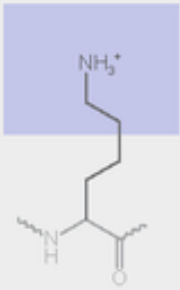
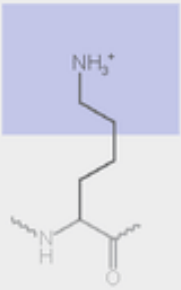
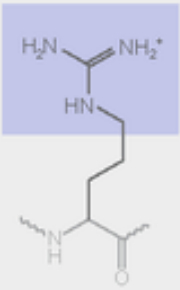
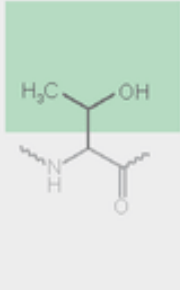


Nonsense variant

Alteration of a single nucleotide results in termination of translation

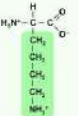
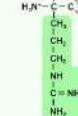
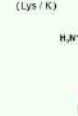




DNA level	TTC	TTT	ATC	TCC	TGC
mRNA level	AAG	AAA	UAG	AGG	ACG
protein level	Lys	Lys	STOP	Arg	Thr
					

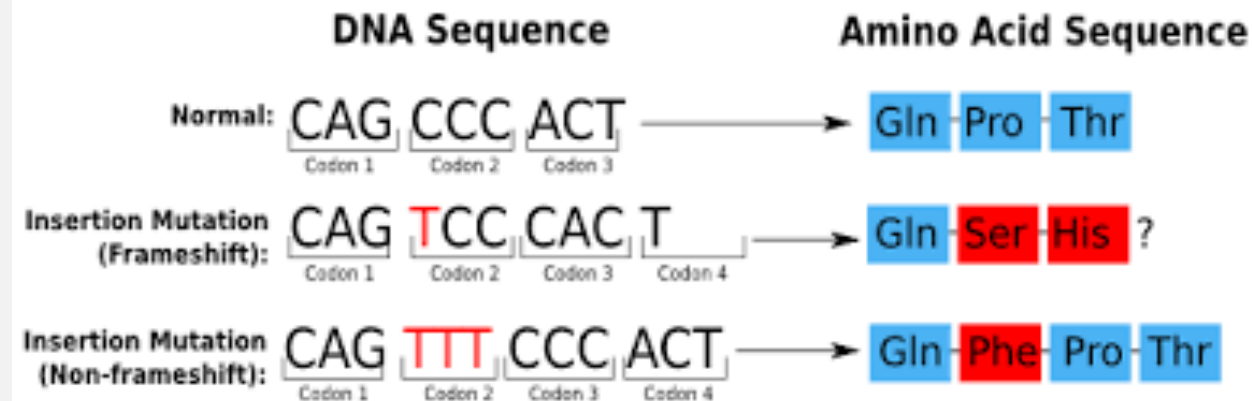
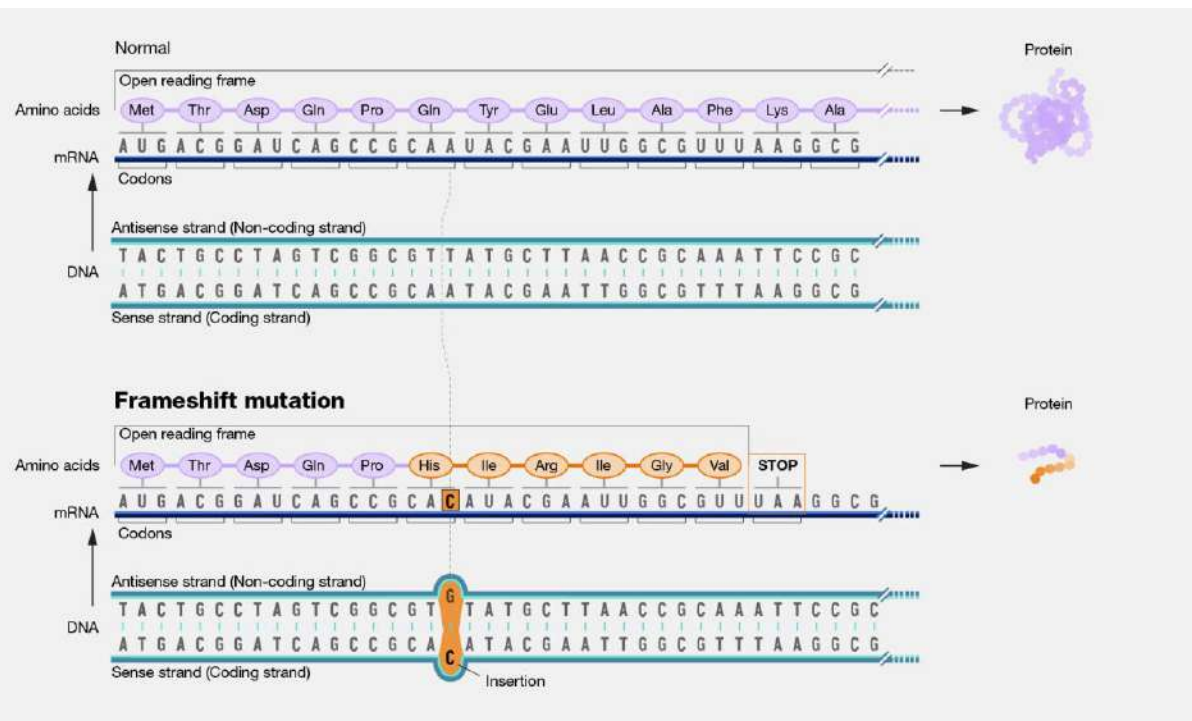
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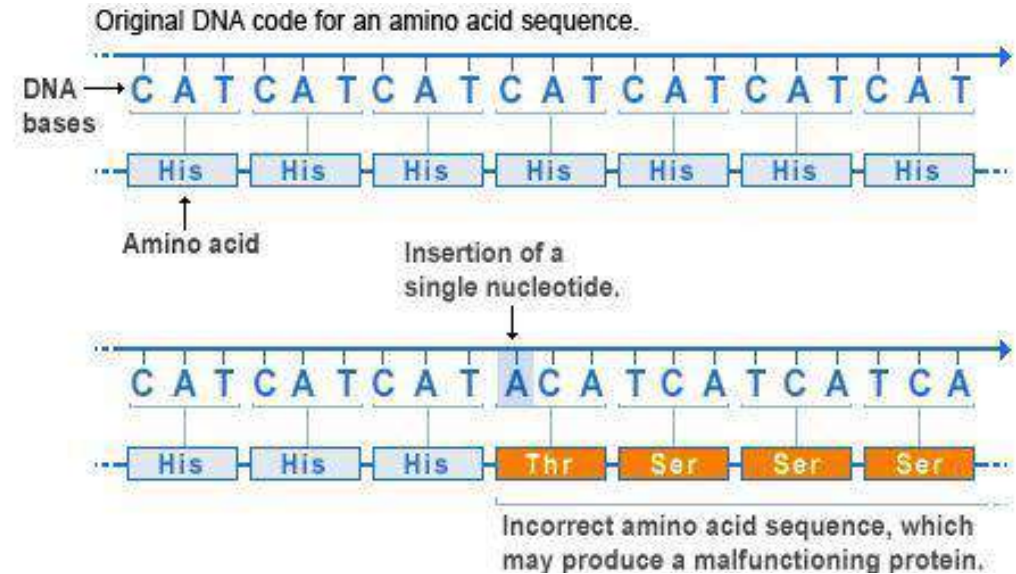
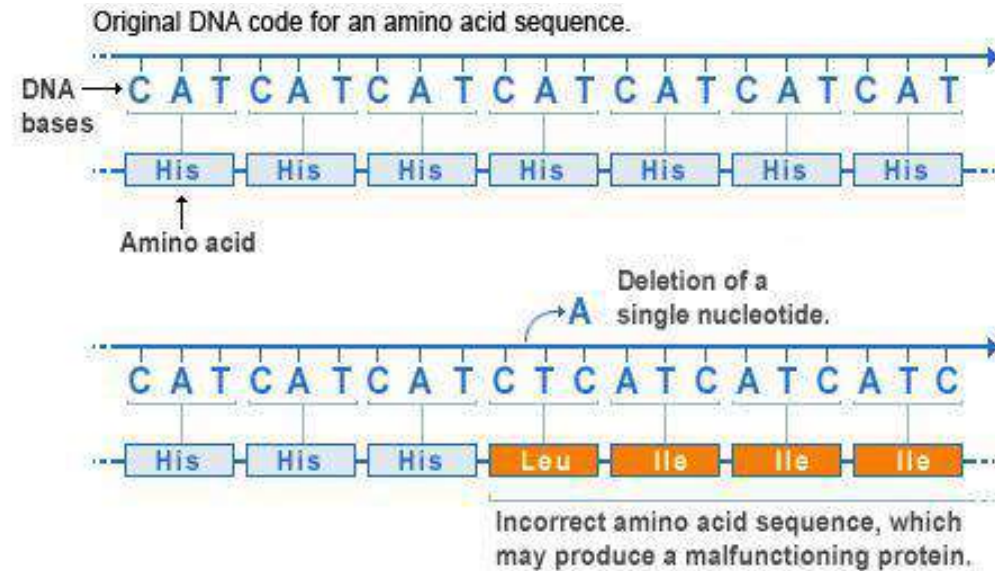
polar

NON-POLAR					+ CHARGE	
 Glycine (Gly / G)	 Alanine (Ala / A)	 Valine (Val / V)	 Cysteine (Cys / C)	 Proline (Pro / P)	 Lysine (Lys / K)	 Arginine (Arg / R)
 Leucine (Leu / L)	 Isoleucine (Ile / I)	 Methionine (Met / M)	 Tryptophan (Trp / W)	 Phenylalanine (Phe / F)	 Histidine (His / H)	
POLAR					- CHARGE	
 Serine (Ser / S)	 Threonine (Thr / T)	 Tyrosine (Tyr / Y)	 Asparagine (Asn / N)	 Glutamine (Gln / Q)	 Aspartic Acid (Asp / D)	 Glutamic Acid (Glu / E)

Frameshift Variant/Mutation

- Insertion or deletion of nucleotides that are **not a multiple of three, it will disrupt the reading frame**
- The amino acid sequence of the protein subsequent to the mutation bears no resemblance to the normal sequence and may have an adverse effect on its function
- Most of frameshift mutations result in a **premature stop codon** downstream to the mutation and may lead to **expression of a truncated protein** unless the mRNA is degraded by nonsense- mediated decay.





Dynamic Variants

- Expansion of trinucleotide repeat sequences
- A number of single-gene disorders have subsequently been shown to be associated with triplet repeat expansions (Table 2.5)
- Described as dynamic mutations because the repeat sequence becomes more unstable as it expands in size

Table 2.5

Examples of diseases arising from repeat expansions

Disease (Gene)	Repeat Sequence	Normal Range (Repeats)	Pathogenic Range (Repeats)	Repeat Location
Huntington disease (<i>HTT</i>)	CAG	9–35	36–100	Coding
Myotonic dystrophy type 1 (<i>DMPK</i>)	CTG	5–35	50–4000	3' UTR
Myotonic dystrophy type 2 (<i>CNBP</i>)	CCTG	11–26	75–>11000	Intron 1
Fragile X site A (<i>FMR1</i>)	CGG	10–50	200–2000	5' UTR
Kennedy disease (<i>AR</i>)	CAG	13–30	40–62	Coding
Spinocerebellar ataxia 1 (<i>ATXN1</i>)	CAG	6–36	39–80	Coding
Spinocerebellar ataxia 2 (<i>ATXN2</i>)	CAG	13–31	32–79	Coding
Machado–Joseph disease/Spinocerebellar ataxia 3 (<i>ATXN3</i>)	CAG	14–44	52–86	Coding
Spinocerebellar ataxia 6 (<i>CACNA1A</i>)	CAG	4–18	19–33	Coding
Spinocerebellar ataxia 7 (<i>ATXN7</i>)	CAG	7–17	38–220	Coding
Spinocerebellar ataxia 8 (<i>ATXN8</i>)	CTG	15–50	71–1300	3' UTR
Spinocerebellar ataxia 10 (<i>ATXN10</i>)	ATTCT	10–29	400–4500	Intron 9
Spinocerebellar ataxia 12 (<i>PPP2R2B</i>)	CAG	7–32	51–78	5' UTR
Spinocerebellar ataxia 17 (<i>TBP</i>)	CAG	25–44	47–63	Coding
Dentatorubral–pallidoluysian atrophy (<i>ATN1</i>)	CAG	7–23	53–88	Coding
Friedreich ataxia (<i>FXN1</i>)	GAA	5–30	70–>1000	Intron 1
Fragile X site E (<i>AFF2</i>)	CCG	6–25	>200	Promoter
Oculopharyngeal muscular dystrophy (<i>PABPN1</i>)	GCG	6	8–13	Coding

UTR, Untranslated region.

Main classes, groups and types of mutations and effects on protein products

Class	Group	Type	Effect
Substitution	Synonymous	Silent	Same amino acid
	Non-synonymous	Missense	Altered amino acid- may affect protein function or stability
		Nonsense	Stop codon- loss of function or expression because of degradation of mRNA
		Splice site	Aberrant splicing- exon skipping or intron retention
		Promoter	Altered gene expression
		Enhancer	Altered gene expression
Deletion	Multiple of three (codon)		In-frame deletion of one or more amino acid(s)—may affect protein function or stability
	Not multiple of three	Frameshift	Likely to result in premature termination with loss of function or expression
	Large deletion	Partial gene deletion	May result in premature termination with loss of function or expression
		Whole gene deletion	Loss of expression
Insertion	Multiple of three (codon)		In-frame insertion of one or more amino acid(s)—may affect protein function or stability
	Not multiple of three	Frameshift	Likely to result in premature termination with loss of function or expression
	Large insertion	Partial gene duplication	May result in premature termination with loss of function or expression
		Whole gene duplication	May have an effect because of increased gene dosage
	Expansion of trinucleotide repeat	Dynamic mutation	Altered gene expression or altered protein stability or function

Variants in Non-coding DNA

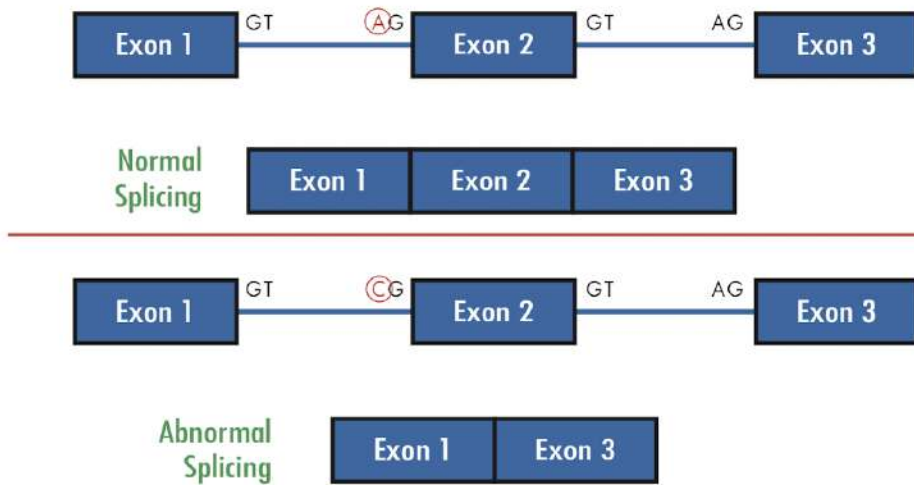
Variants in Non-Coding DNA

Splice Variant

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graph TD; A[Variants in Non-Coding DNA] --- B[Splice Variant]
```

Splice Variants

Splice Site Mutation



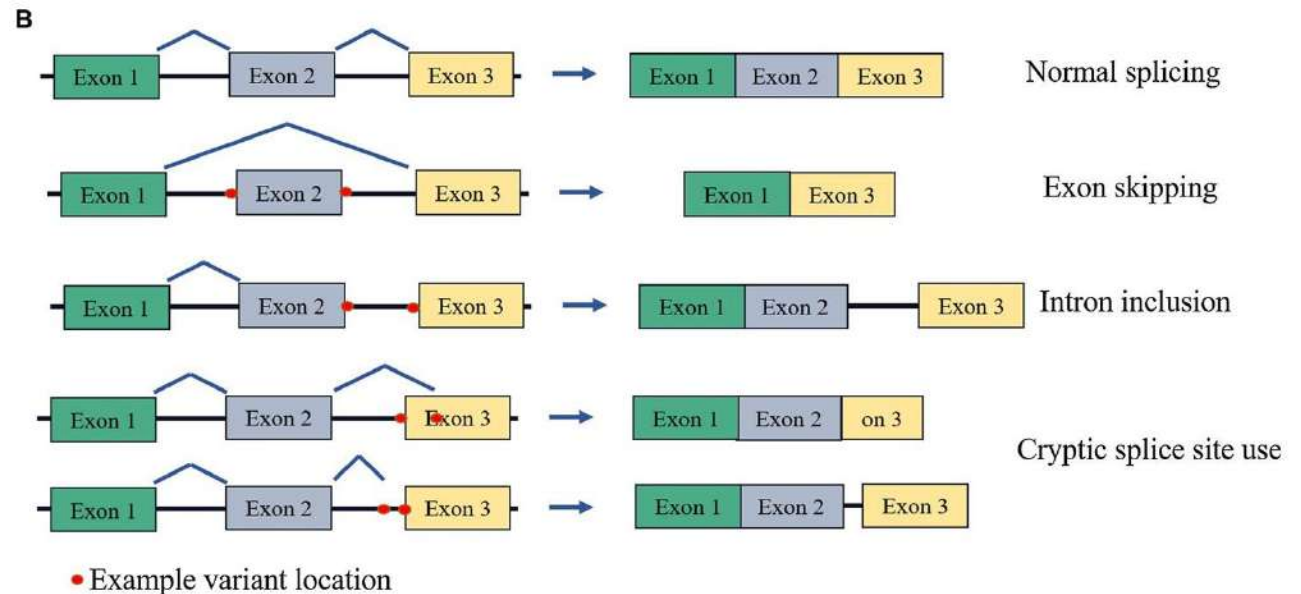
Definition:

- A genetic alteration in the DNA sequence that occurs at **intron-exon boundaries (splice site)** which alter the splicing signals that is important for excision of introns
- It occur in **mature RNA** only
- Can occur at splice donor (GT) and splice acceptor (AG) sites, usually result in aberrant splicing

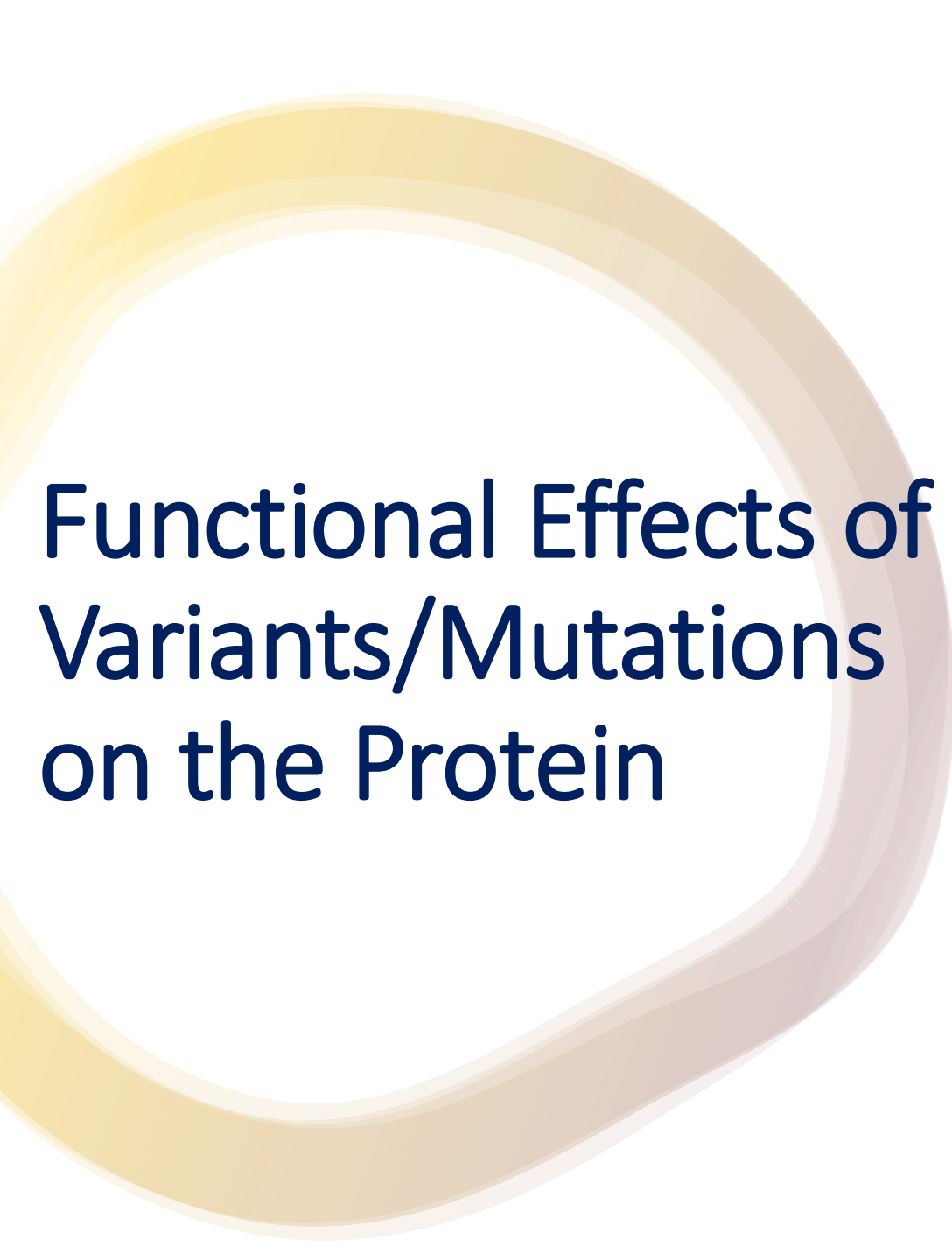


Consequence or effect: Splice Variants

- Can result in loss of exon (**exon skipping**) or **retention of intron**
→ may lead to frameshift mutations
- **Cryptic splice sites**, which resemble the sequence of an authentic splice site, may be activated when the conserved splice sites are mutated.
- Cryptic site for splicing results in partial deletion of the exon or the deletion of an entire exon



Lord, J., & Baralle, D. (2021). Splicing in the diagnosis of rare disease: advances and challenges. *Frontiers in Genetics*, 12, 689892.

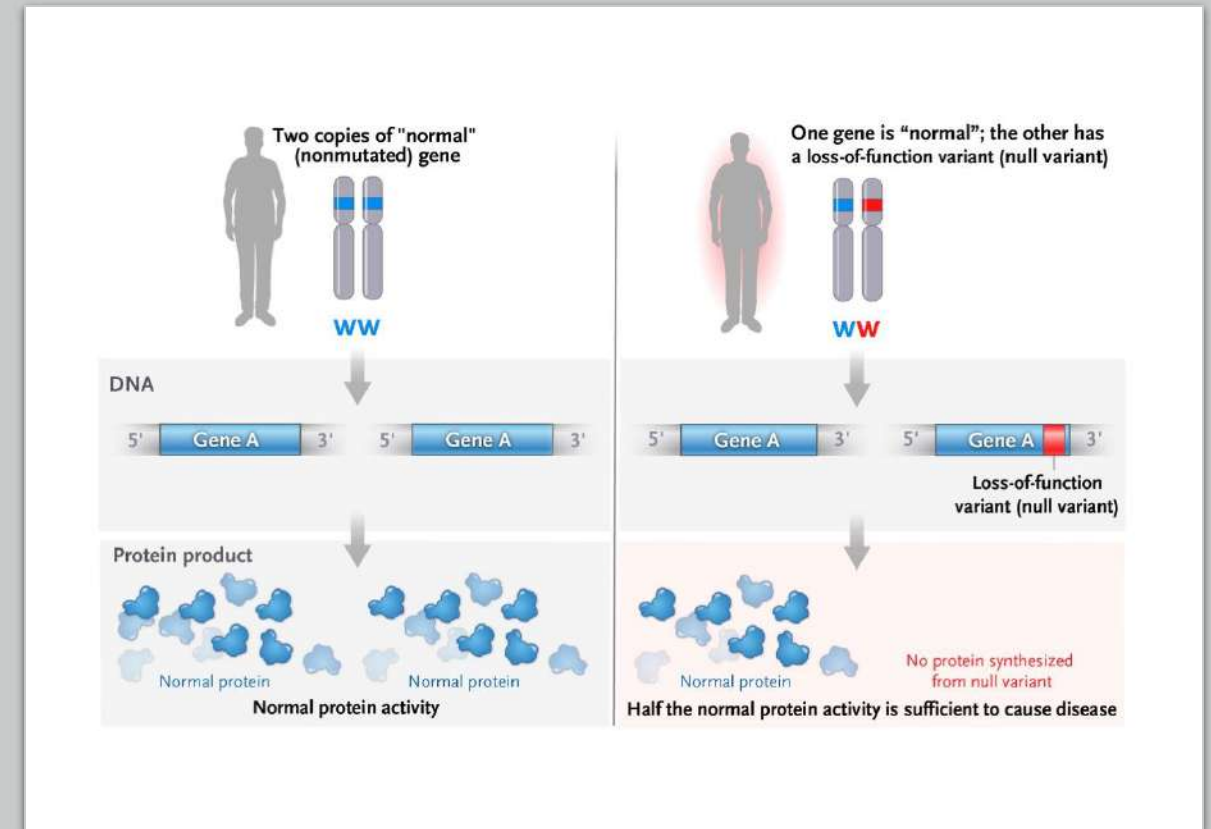
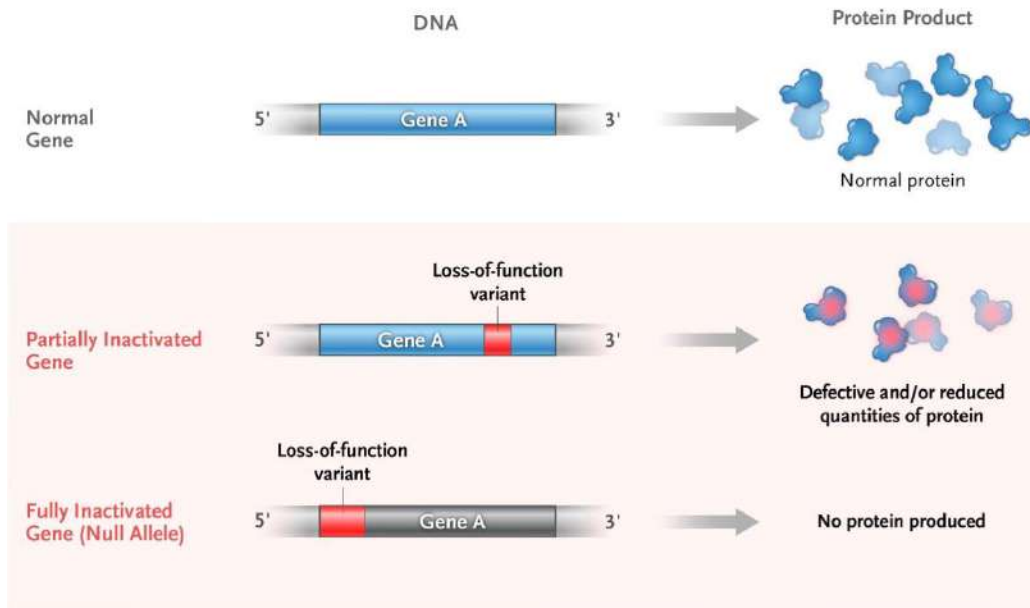


Functional Effects of Variants/Mutations on the Protein

- Complete or partial Loss-of-function (LOF) variants (amorph/hypomorph)
- Gain-of-function variants
- Dominant-negative variants

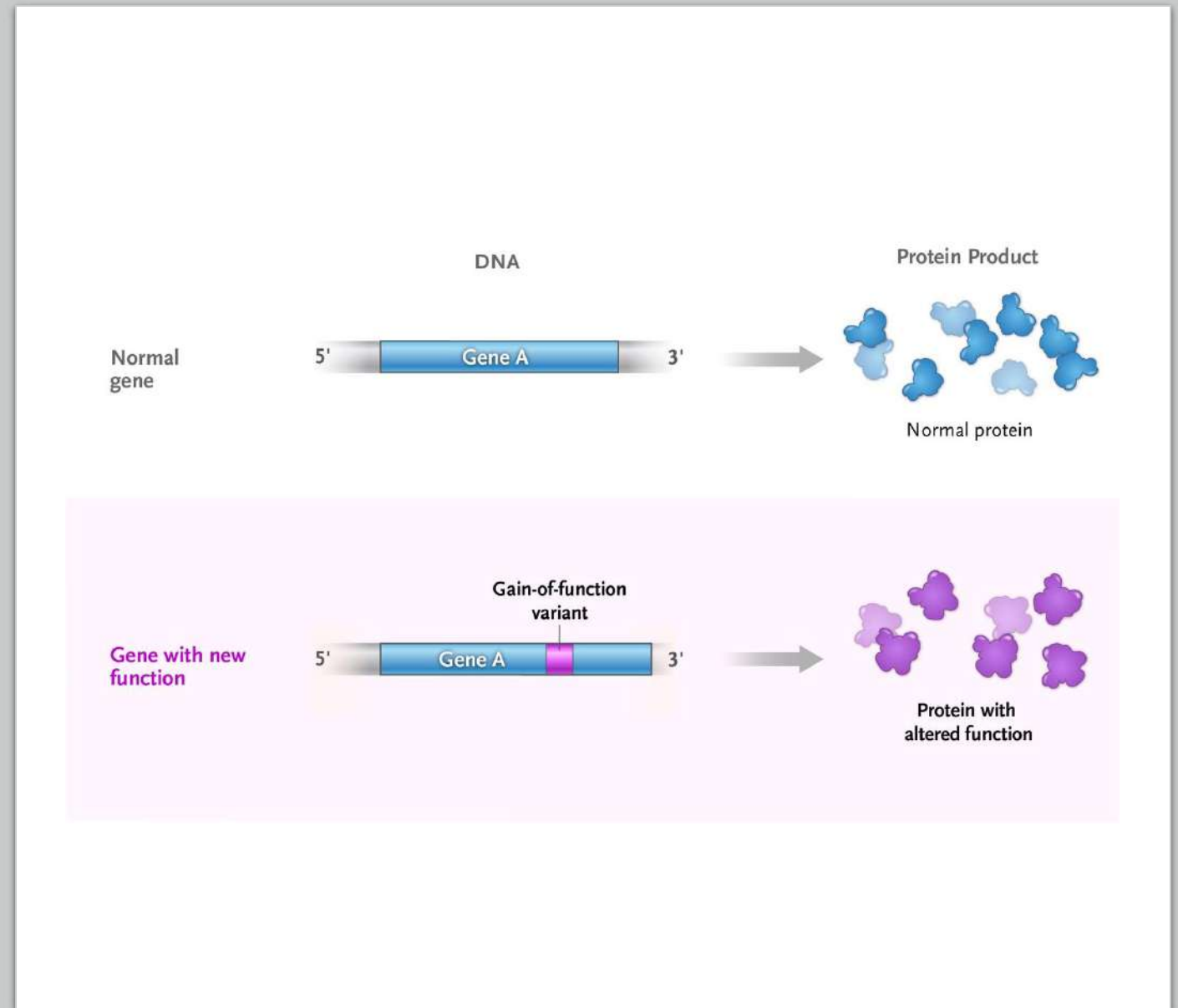
Loss-of-Function (LOF) Variants:

- Can result in either reduced activity or complete loss of the gene product
- **Haplo-insufficiency variants:** LOF variants in the heterozygous state in which half normal levels of the gene product result in disease
- E.g., familial hypercholesterolemia



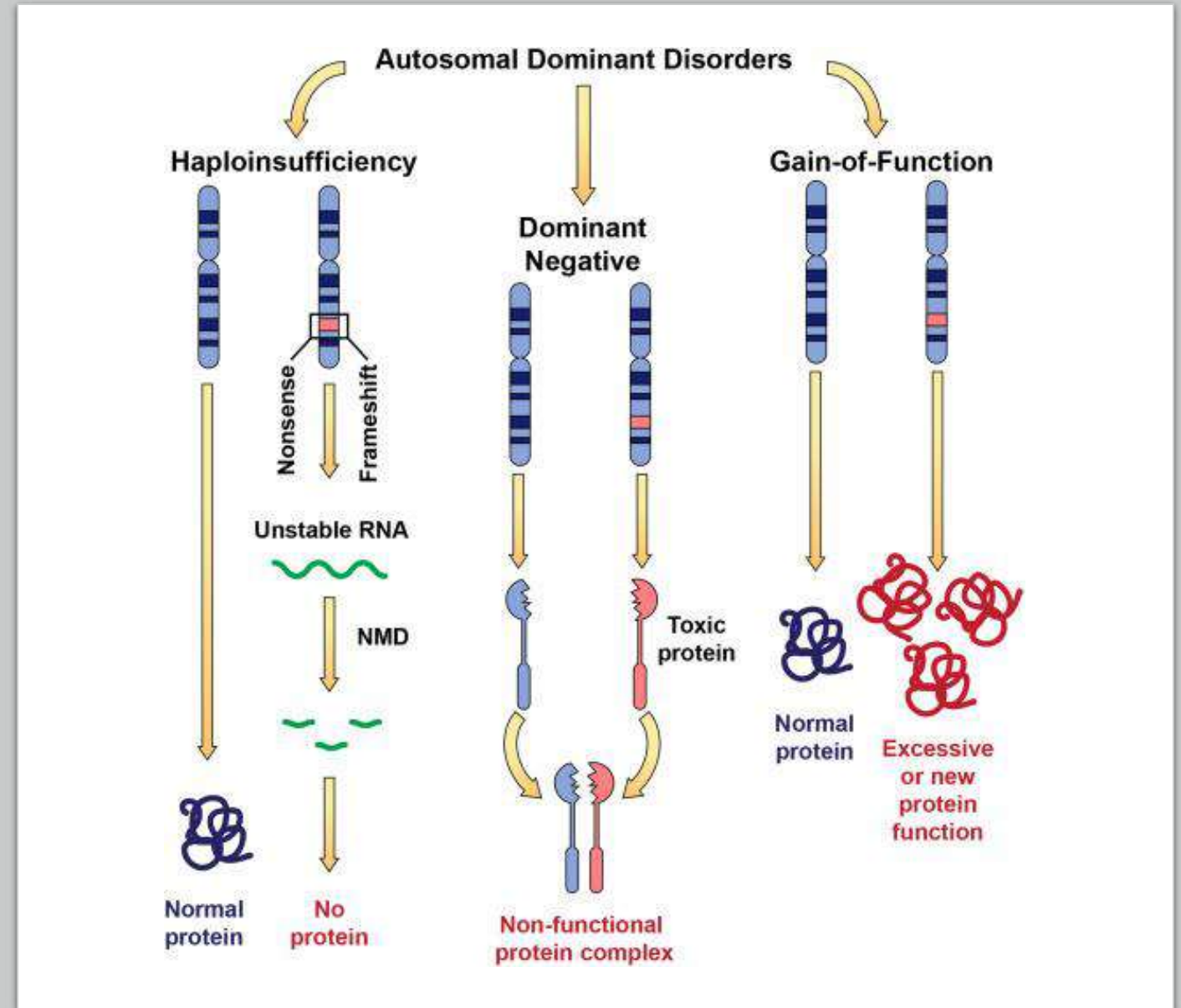
Gain-of-Function Variants:

- Result in either increased levels of gene expression or the development of a new function of the gene product
- E.g., Huntington disease and achondroplasia

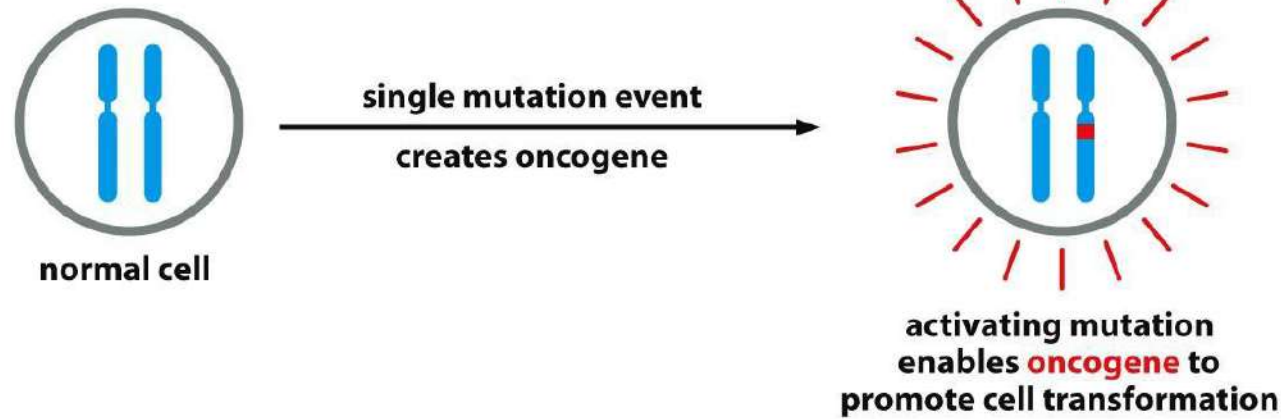


Dominant-Negative Variants:

- When a mutant gene in the heterozygous state interferes with the function of the normal gene product of the corresponding allele
 - results in loss of protein activity or function
- E.g., osteogenesis imperfecta and Marfan syndrome



(A) **overactivity mutation** (gain of function)



cells
en route to
cancer

(B) **underactivity mutation** (loss of function)

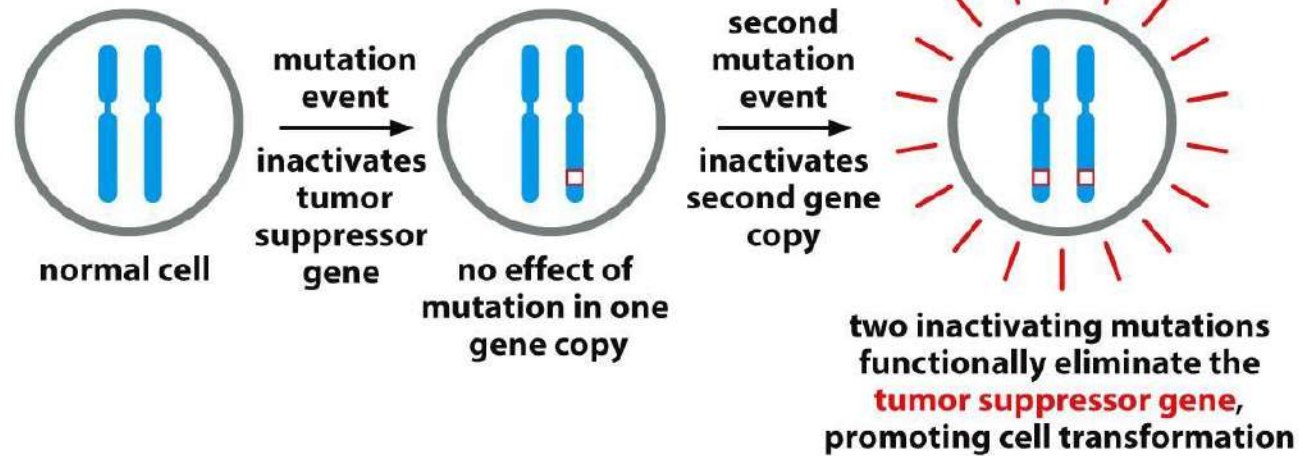


Figure 20-17 Molecular Biology of the Cell 6e (© Garland Science 2015)

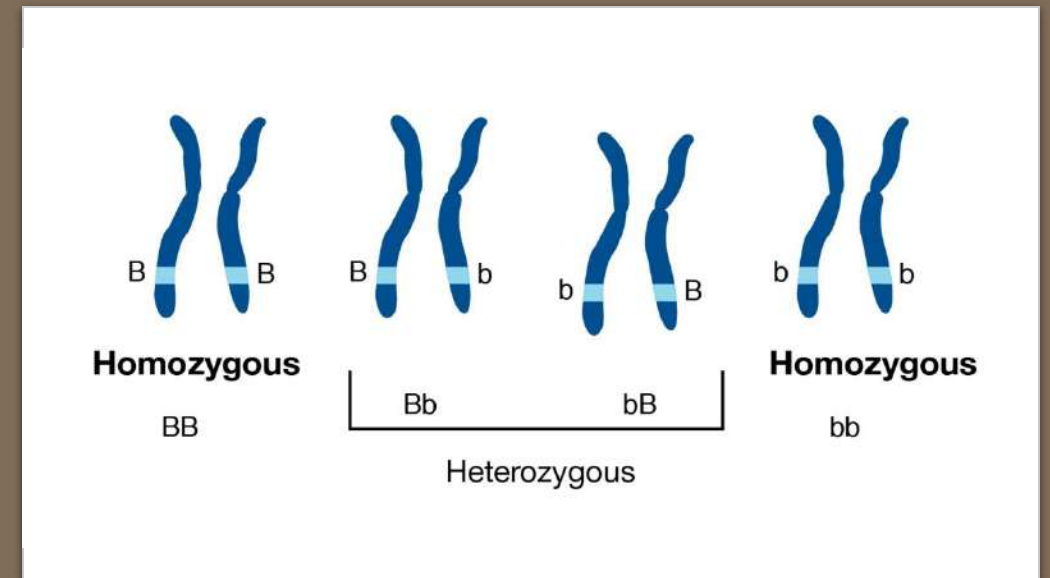
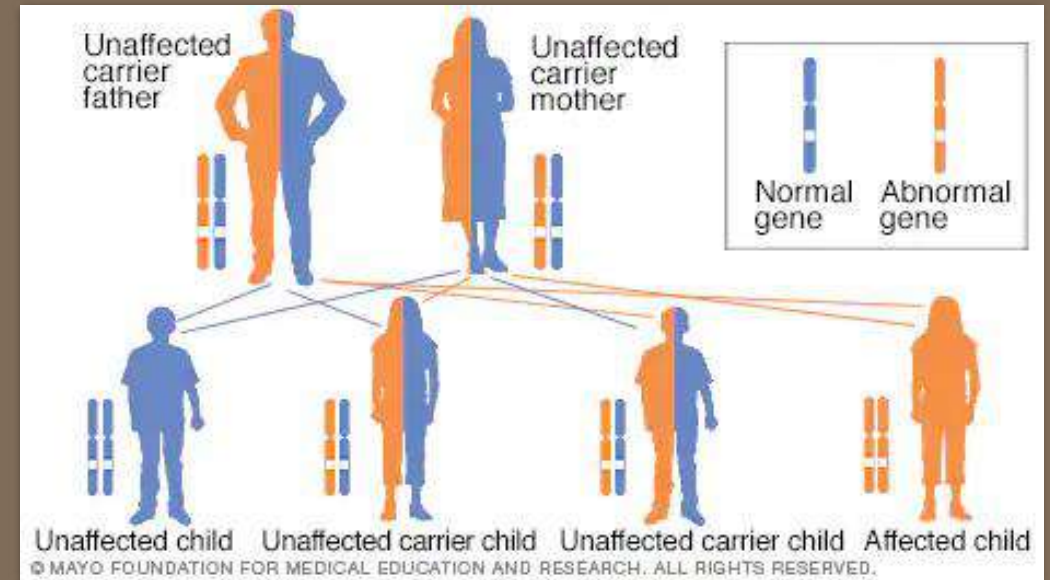
Variants/Mutations

– By inheritance

- Homozygous variants
- Compound heterozygous (CH) variants
- *De Novo* variants

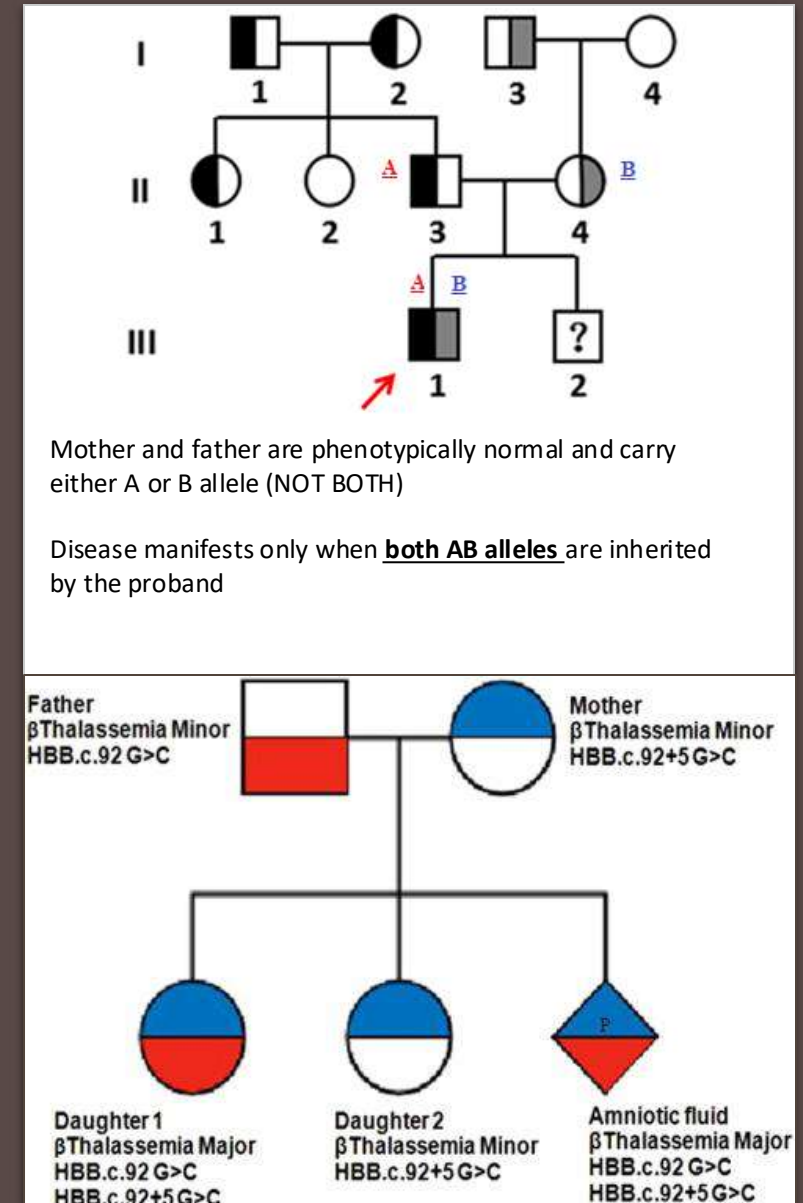
Homozygous Variants

- Occur when an individual has **two identical copies of a mutation** in a particular gene (two same variants in the same gene)
- Both copies of homozygous variants show same deleterious effect on protein's structure and function
- Causes different autosomal recessive disorders
- These disorders are usually passed on by two carriers
- Their health is rarely affected, but they have one mutated gene and one normal gene



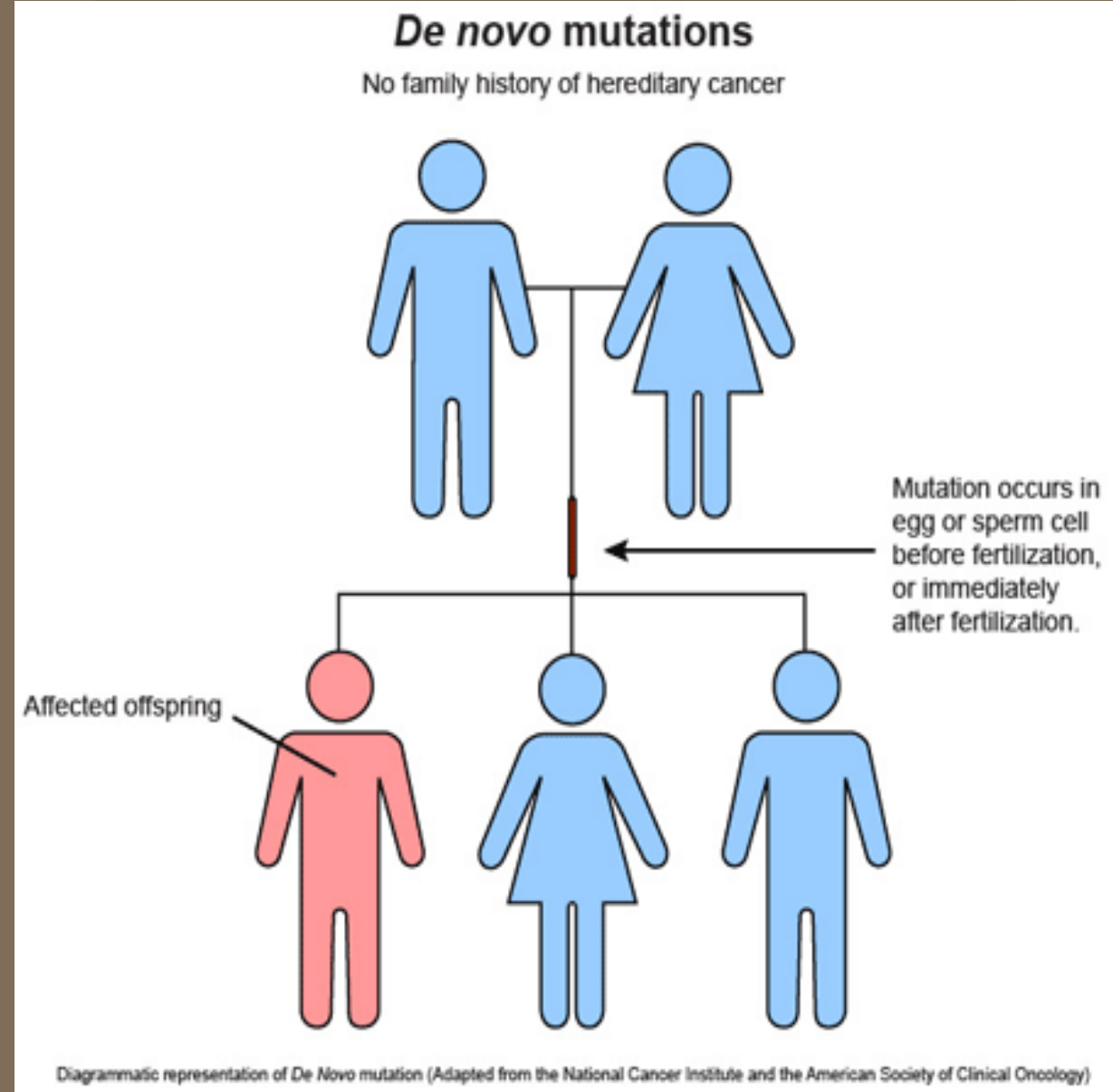
Compound Heterozygous (CH) Variants

- CH variant occurs when a person inherits two heterozygous alleles, one from each parent, and these alleles are located at different positions within the same gene (two different variants in the same gene)
- These heterozygous alleles produce a structurally and/or functionally defective protein
- Cause autosomal recessive disorders



De Novo Variants

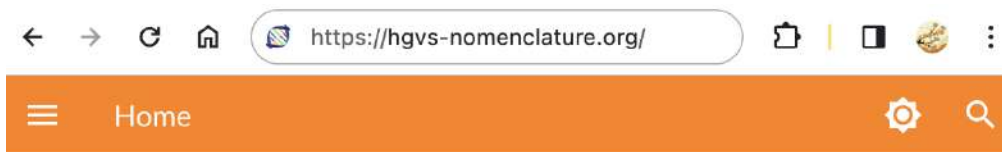
- A *de novo* variant is a **new mutation** that occurs in a **germ cell** (egg or sperm) of one of the parents, then passed on to an offspring
- Associated with advanced paternal age, as consequence of the large number of mitotic divisions that male gamete stem cells undergo during a man's reproductive lifetime
- Produce deleterious effect on protein's structure and function



Genotype– Phenotype Correlation



- Is to correlate the presence of a particular mutation (**genotype**) with the specific features seen in a person with disorder (**phenotype**)
- **Example:** variants in the receptor tyrosine kinase gene RET which, depending on their location, can lead to different syndromes that differ in functional mechanism and clinical phenotype
 - Loss-of-function nonsense variants → lack of migration of neural crest–derived cells to form the ganglia of the myenteric plexus of the large bowel → Hirschsprung disease
 - Gain-of-function missense variants → familial medullary thyroid carcinoma or multiple endocrine neoplasia type 2



Human Genome Variation Society (HGVS)

The **HGVS Nomenclature** is an internationally-recognized standard for the description of DNA, RNA and protein sequence variants. It is used to convey variants in clinical reports and to share variants in publications and databases.

The HGVS Nomenclature is administered by the [HGVS Variant Nomenclature Committee \(HVNC\)](#) under the auspices of the [Human Genome Organization \(HUGO\)](#).

- The most important rule is that all variants should be described at the **most basic level**, i.e., the DNA level
- Descriptions should always be in relation to a **reference sequence**, either a genomic or a coding DNA reference sequence
- All changes reported must be described at DNA-level
- When descriptions at RNA or protein level are given in the text between ()

Understand Nomenclature of Genetic Variations

- "c." for a coding DNA sequence (like c.76A>T)
- "g." for a genomic sequence (like g.476A>T)
- "m." for a mitochondrial sequence (like m.8993T>C)
- "n" for a non-coding RNA reference sequence (gene producing an RNA transcript but not a protein)
- "r." for an RNA sequence (like r.76A>U)
- "p." for a protein sequence (like p.Lys76Asn)

F5 NM_000130.3: c. 1601G>A (p.Arg534Gln)

↑
HGNC
Official
Gene
Symbol

↑
Reference
sequence
(Genbank
accession
No. and
Version No.)

↑
Coding
DNA
sequence

↑
Guanine-to-
Adenine
at position
1601

↑
"Protein"

↑
Arg-to-
Gln
at codon
534

Table 2.4 Mutation nomenclature: examples of CFTR gene mutations

Type of Mutation	Nucleotide (Ref Seq NM_000492.3)	Protein Designation	Consequence Description
	c.350G>A	p.Arg117His	
	c.1624G>T	p.Gly542*	
	c.489 + 1G>T		
	c.948delT	p.Phe316Leufs*12	
	c.1521_1523delCTT	p.Phe508del	
	c.3767dupC	p.Leu1258Phefs*7	

Mutations can be designated according to the genomic or complementary DNA (mRNA) sequence and are prefixed by “g.” or “c.,” respectively. The first base of the start codon (ATG) is c.1.

References

- Emery's Elements of Medical Genetics 16th edition 2022 by Turnpenny and Ellard
- <https://www.genome.gov/genetic-s-glossary/>
- <https://illustrated-glossary.nejm.org/>